Supplementary Material

Hollingham (2026) — Newborough Warren Coastal Sand Dune Aquifer, Isle of Anglesey

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Supplementary Note S1: Interactive Map

Supplementary File S1: Interactive map of the full dipwell monitoring network (Google Maps), accessible at:
https://www.google.com/maps/d/viewer?mid=1hXLAauiMeaVsXhBR_IoUTziAtjk

Supplementary Note S2: Aquifer Geometry, Hydraulic Parameterisation and the Basis of the Spatial Flow Model

S2.1 The Borehole Data Problem

Darcy's law and any spatially explicit groundwater flow model require three spatially resolved inputs: hydraulic conductivity (K), aquifer thickness (b, from which transmissivity T = K·b is derived), and hydraulic gradient (∇h). For the Newborough Warren dune system, only the third of these — the hydraulic gradient — is well constrained, through the 88-dipwell monitoring network. Hydraulic conductivity and aquifer thickness are poorly constrained, because borehole logs providing lithological or geophysical evidence of the depth to the bedrock or glacial till substrate are absent across the great majority of the 700 ha dune system.

The available borehole evidence is confined to four locations documented in Betson et al. (2002), a report commissioned by the Countryside Council for Wales (Contract FC†73-05-18) covering hydrogeological investigations at the site. These boreholes provide minimum depth-to-basement constraints at four spatial points and constitute the only direct subsurface lithological evidence available for the site. They are summarised in Table S2.1.

Borehole	Easting (OSGB36)	Northing (OSGB36)	Sat. thickness (m)	Source
Water borehole	241,721	364,133	12.8 (min.)	Betson et al. (2002)
NH1	241,734	363,306	6.5	Betson et al. (2002)
NH2	242,024	363,107	6.5	Betson et al. (2002)
Borehole 3	242,837	363,288	3.65	Betson et al. (2002)

Table S2.1. *Borehole constraints on saturated thickness from Betson et al. (2002). Values represent confirmed minimum depth to the low-permeability substrate (glacial till or bedrock). The Water borehole value of 12.8 m is a minimum — drilling did not reach the substrate. The eastern borehole (Borehole 3) shows progressive thinning toward the Menai Strait. These four boreholes constitute the only direct subsurface lithological evidence available for the 700 ha dune system and represent the most important data gap for any future spatially explicit groundwater model.*

The tracer test reported by Betson et al. (2002) provides the single direct estimate of hydraulic conductivity for the site, yielding K = 6.0 m/day at the central dune plain (range 3.0–9.0 m/day from the sensitivity analysis). This value is used as the site-wide K constant in the drawdown-propagation construction. It does NOT enter the flow arrows of §4.9.5 (Figure 56), which are the kriged head gradient and are independent of both K and aquifer thickness — as stated four paragraphs below. The range provides the basis for the exploratory uncertainty envelope noted in the figure annotations.

S2.2 The β_1 Proxy for Relative Aquifer Thickness

The physical interpretation of β_1 is the fraction of a unit rainfall pulse that reaches the water table as a head rise, integrated over the monthly timestep. In an unconfined aquifer, this is governed by the depth of the unsaturated zone and the specific yield: a thick unsaturated zone above a large storage volume attenuates the rainfall-to-head transfer, producing a low β_1. A shallow, low-storage aquifer (thin unsaturated zone, low Sy) produces a high β_1 — rainfall arrives at the water table quickly and raises it sharply. Although β_1 cannot be inverted algebraically to yield aquifer thickness without independent knowledge of the recharge flux and specific yield, its spatial pattern is a reliable proxy for the relative contrast in aquifer thickness across the site.

The Eastern Block (C1: β_1 = 4.577; C2: β_1 = 3.989) and Western Residual cluster (C3: β_1 = 3.808) contrast is consistent with the geological interpretation of Stratford et al. (2007): the Eastern Block sits on shallow till and estuarine deposits, producing a thin, storage-limited aquifer with a flashy response; the Western Residual Cluster occupies deep, clean aeolian sand, producing a capacious, buffered aquifer with an attenuated response. The Forest cluster (C4: β_1 = 2.502) is anomalously low even relative to C3, reflecting canopy interception rather than additional aquifer depth — consistent with the depth-dependent PET analysis (Script 15), which returns the weakest depth coupling in the network at C4 (best κ = 0.05 m^{-1}, against 2.30 m^{-1} at C1), indicating that capillary disconnection at depth is not the primary control. C5 (Coastal Forest: β_1 = 2.462) shows a comparable value to C4, consistent with both clusters carrying the same Corsican pine canopy on the same deep sand substrate.

Cluster	Label	β_1	β_2	β_3	Geological context
C1	Lake Edge	4.577	0.942	0.088	Shallow till/estuarine substrate (NH1, NH2 boreholes: 6.5 m); rapid lake exchange
C2	Dune	3.989	1.842	0.062	Shallow till substrate consistent with C1; mature open dune
C3	Western Residual	3.808	1.833	0.062	Deep aeolian sand; Water borehole ≥12.8 m; DEM ridge geometry
C4	Main Forest	2.502	2.564	0.019	Same deep sand substrate as C3; low β_1 reflects 24% canopy interception
C5	Coastal Forest	2.462	1.336	0.045	Pine canopy on coastal sand; geomorphological thinning toward Menai Strait

Table S2.2. *Cluster mechanistic coefficients (from Table 3, displacement-formulation SSM) and geological context. The β_1 contrast between Eastern Block (C1/C2) and Western Residual cluster (C3) is consistent with the stratigraphic interpretation of Stratford et al. (2007). C3 and C4 share the same deep aeolian sand body, confirmed by β_1 similarity after*

correcting for canopy interception and by Pearson affinity persistence post-felling. Borehole 3 (3.65 m, eastern margin) confirms progressive thinning toward the coast.

S2.2.1 Aquifer Thickness Surface (Developed for the PDE Model)

During development, a two-dimensional finite-difference partial differential equation (PDE) solver was implemented as an alternative spatial framework (see S3.3). That model required a spatially continuous aquifer thickness surface to compute transmissivity $T = K \cdot b$ at each grid cell. Because only four borehole constraints were available, an IDW-interpolated thickness surface was constructed using the borehole values as hard nodes supplemented by cluster-level thickness priors derived from the β_1 proxy argument above and per-well overrides at geological boundaries. The surface was interpolated on a 50 m grid using power-1 IDW with a physical minimum of 0.3 m enforced.

The PDE model was subsequently evaluated and rejected (S3.3) because it produced near-zero site-wide differences for forest management scenarios, primarily due to dilution of the modified β coefficients during grid interpolation, weak drainage feedback at C4, and the absence of ridge-derived lateral recharge from the source terms. The thickness surface is therefore not used in any published result. The per-well equilibrium framework adopted in the main paper (Section 3.8) operates directly from the fitted β coefficients at each well and does not require a thickness parameterisation. The flow arrows shown in Figure 56 are head-gradient arrows from the kriged mean water table and are independent of aquifer thickness.

The borehole constraints in Table S2.1 and the geological contrasts summarised in Table S2.2 remain relevant for any future spatially explicit model. Slug tests at two to three representative wells per cluster and a ground-penetrating radar survey of aquifer thickness are identified as the highest-priority data gaps for future field campaigns (§5.9).

S2.2.2 Per-Well Geological Constraints

Several wells occupy geological settings that depart from their cluster's typical substrate. These are noted here for reference, as they would serve as boundary constraints for any future spatial model.

CEH14 (241,292 E, 364,488 N), classified as C4, sits at the crest of the bedrock ridge at 14.4 m AOD with a mean water table head of 13.3 m AOD. The water table is approximately 1.1 m below ground surface here. CEH14 overlies irregular bedrock topography on the elevated ridge flank, where buried ridges impede lateral drainage — consistent with its anomalous negative β_3. That coefficient is not significant, however ($\beta_3 = -0.016$, $p = 0.24$), and six of the nine C4 wells likewise carry a non-significant β_3 on the hundred-month per-well window, so CEH14 sits at the imprecise end of a cluster that is imprecisely resolved on that window; the negative sign should not be read as a result in its own right. Refitting each well on its full record leaves seven of the nine significant, with CEH13 and CEH14 the two that still fail, so the imprecision is largely a property of the window rather than of the cluster. Its water-balance residual is −0.010 m/month, the most negative in the network (§4.9.6, Figure 59).

Wells CEH7 (243,386 E, 363,613 N) and CEH8 (243,150 E, 363,382 N) sit at the far eastern estuarine margin of the site, where the DEM confirms ground elevation of 1.2–5.2 m AOD. Both represent coastal pinch-out positions where the aquifer thins to approximately 4.0 m. and their records are intermittent.

S2.3 Why C4 Retains Its Cluster Identity After Clearfell

A key question for the scenario analysis is whether the clearfell treatment wells (FE1, FE2, FE4, LIS1) converged toward C3 open-dune behaviour after the December 2017 felling. The Pearson affinity analysis (Section 4.3) provides a direct empirical test: all four impact wells maintained r > 0.97 affinity with C4 even in the post-felling period (2018–2026), with no significant shift toward C3 affinity.

This persistence is consistent with the interpretation that the Forest cluster signature at these wells reflects deep sandy substrate at high topographic elevation rather than canopy influence alone — in which case convergence toward open-dune behaviour may not be achievable regardless of management intervention. C4 wells sit at the highest topographic positions in the dune system (mean head 9.24 m AOD), and their deep unsaturated zones and low recharge sensitivity reflect their geological position, not only their canopy cover. This conclusion is independently supported by the NW10 broadleaf comparison (§5.6.3): NW10, situated within the 1993 clearfell and 1995 broadleaf restocking block, has maintained C4 cluster affinity (r = 0.986) throughout the 18-year monitoring period despite its different canopy cover, confirming that substrate position rather than canopy type is the dominant control on cluster identity.

S2.4 Reproducibility

The spatial model parameterisation documented in this note is implemented in 19_spatial_groundwater.py (scenario viewer data preparation, IDW thickness surface, per-well equilibrium calculations) and 20_spatial_figures.py (flow-direction quiver — a unit-normalised head gradient, not a Darcy flux — mean head surface, water balance residual field). The β_1 proxy analysis draws on the cluster-level SSM coefficients exported by 03_state_space_model.py (03_03_cluster_mechanistic_coefficients.csv). The Pearson affinity test for C4 cluster persistence post-felling is computed in 05_pearson_affinity.py. All scripts read intermediate data produced by the main pipeline and are maintained in the canonical pipeline sequence. The interactive scenario viewer (scenario_viewer.html) is a standalone browser application that reads the per-well coefficient and location CSVs exported by Script 19.

Supplementary Note S3: Scenario Modelling Framework and Limitations

S3.1 Purpose and Scope of the Scenario Analysis

An exploratory scenario analysis was developed to investigate whether the site's own SSM β coefficients could yield defensible quantitative predictions of water table response to management interventions and climate change. This note documents the scenario framework adopted (the per-well equilibrium framework described in Section 3.8), the alternative PDE-based approach that was evaluated and rejected during development, and the structural limitations that apply to any equilibrium representation of the system. The scenario outputs themselves are reported in Section 4.13 and discussed in Section 5.5.2 of the main paper; this note provides the technical reasoning behind the methodological choices and their limitations.

S3.2 The Per-Well Equilibrium Framework

The scenario framework adopted in this study is a well-level equilibrium calculation that applies scenario-specific perturbations to the SSM forcing terms (rainfall, PET, canopy

interception, atmospheric draw) and computes the resulting change in equilibrium head for each reference well. Per-well Δh values are then interpolated to a 50 m grid on the British National Grid by Delaunay triangulation with linear barycentric weighting (scipy.interpolate.griddata, method="linear"), masked to the rectangular sea-boundary extent; the resulting field is rendered in the interactive scenario viewer rather than as a static map. The accompanying interactive scenario viewer (scenario_viewer.html) renders the same per-well Δh field using power-1 eight-nearest-neighbour inverse-distance weighting for responsiveness under slider-driven re-interpolation, and additionally offers a depth-below-surface view (DEM minus interpolated head, per cell) with dune-ridge cells masked — see S3.7 for the viewer-specific rendering conventions. The full per-scenario × per-season × per-cluster Δh output is tabulated in the supplementary file 19_scenario_summary.csv.

The scenario head change for each well is computed directly from the fitted SSM equation as

$$\Delta h = (\beta_1 \cdot P_eff,sc - \beta_2,sc \cdot PET_sc - \beta_3 \cdot \hbar_disp) - (\beta_1 \cdot P_eff,0 - \beta_2 \cdot PET_0 - \beta_3 \cdot \hbar_disp)$$

where the $\beta_3 \cdot \hbar$_disp term cancels algebraically between scenario and baseline expressions, so β_3 does not materially affect the computed Δh. Specific yield does not appear in the equation because the fitted β coefficients already embed the aquifer storage response through the SSM fit to observed head data; the per-well WTF-derived Sy values (Section 3.7.3) are retained as supplementary display information in the interactive viewer, not as a divisor on Δh. The framework is honest about what it is: a geometric aggregator of SSM responses across the cluster structure, not a physical model of groundwater flow.

The per-well framework has two important structural features that shape the interpretation of its outputs. First, scenario parameter changes for canopy interception and β_2 apply only to the C4 and C5 Forest cluster wells, because that is where the relevant physical changes occur under forest management. The framework therefore necessarily produces a response confined to the C4/C5 zone and its immediate interpolated surroundings, with effectively no propagation into C1, C2, or wider C3 clusters. Second, climate scenario parameter changes (ΔP, ΔPET) apply to all reference wells simultaneously, so the resulting Δh field is spatially continuous and reflects the site-wide β coefficient structure. The contrast between forest management and climate scenarios in the spatial outputs is therefore a direct consequence of which wells are affected by which perturbations, not an artefact of the numerical method.

S3.3 Evaluated-and-Rejected: The 2D Steady-State PDE Model

During development, an alternative spatial framework was implemented as a two-dimensional finite-difference PDE solver based on the Helmholtz form of the steady-state groundwater flow equation:

$$\nabla \cdot (T \cdot \nabla h) - \beta_3 \cdot h = -(\beta_1 \cdot P_eff - \beta_2 \cdot PET)$$

with transmissivity T = K·b (K = 6.0 m/day after Betson et al., 2002; b IDW-interpolated from four borehole constraints supplemented by cluster-level priors derived from the β_1 proxy, also after Betson et al., 2002). Dirichlet head conditions (h = 0 m AOD) were applied at sea boundaries, and implicit Neumann no-flow conditions at the ridge. The sparse linear system of approximately 4,200 equations was solved using scipy.sparse.linalg.spsolve (Virtanen et al., 2020).

This approach is well-posed mathematically — the Helmholtz structure guarantees a unique solution with the given boundary conditions — and was physically consistent in that the β coefficients entering the source term are the same parameters estimated from the observed well records. The solver produced physically plausible baseline head surfaces and Darcy flux fields qualitatively similar to those obtained from the per-well framework.

However, the PDE solver produced near-zero site-wide differences for forest management scenarios, for three structural reasons.

First, the forest management scenarios modify β coefficients only for the C4 and C5 Forest wells. These per-well β values are then IDW-interpolated to the 4,235-cell grid. Because the forest wells constitute a small minority of the 88-dipwell network, the modified β values are heavily diluted by the surrounding clusters during interpolation. The source term $\beta_1 \cdot P_eff - \beta_2 \cdot PET$ therefore changes by only a small amount at each cell, and the resulting change in the steady-state head solution is proportionally small.

Second, the internal drainage term $\beta_3 \cdot h$ provides a stabilising feedback in the Helmholtz equation. C4 has the lowest β_3 in the network (0.018, compared with 0.057 for C3 and 0.064 for C2; Table 3). This means that in the C4 zone the drainage feedback is weak, and a moderate source term change produces only a small equilibrium head change.

Third, and most fundamentally, the steady-state PDE does not represent the elevated boundary condition at the ridge margin. The observed mean head at C4 (9.24 m AOD) is substantially higher than the mean at C3 (7.00 m AOD), and this contrast is not a product of C4's β coefficients: the C4 wells occupy the highest topographic positions in the dune system, on and against the bedrock ridge flank. The PDE source terms are parameterised from the SSM β coefficients alone and carry no representation of that boundary, so the solution understates the C4 water table under both baseline and scenario conditions and the difference between the two solutions is correspondingly small. The argument rests on topographic position, which is directly observed. It does not rest on a ridge-derived lateral flux: the corrected water-balance residual field (§4.9.6, Figure 59) provides no evidence for one, since CEH14 — the well most proximal to the ridge — carries the most negative residual in the network (−0.010 m/month), no well exceeds +0.02 m/month, and residual magnitude is uncorrelated with position on either axis (Spearman $\rho = -0.14$ on easting, $p = 0.27$; $\rho = +0.16$ on northing, $p = 0.20$). Whether any lateral flux exists at all cannot be resolved from the water-level record, and the rejection of the PDE approach does not depend on it.

A fully calibrated continuous-flow model would require, at minimum: (i) slug tests at two to three representative wells per cluster to constrain the spatial distribution of K (currently constrained only by the single tracer test of Betson et al., 2002); and (ii) a ground-penetrating radar survey of aquifer thickness, with coring, to replace the indirect β_1-proxy constraints with direct measurements of thickness and specific yield. These are the same parameter-sparsity constraints that produced calibration difficulties in the Betson et al. (2002) MODFLOW model at this site.

For these reasons, the PDE approach was judged to overclaim relative to the data available and was rejected in favour of the per-well equilibrium framework. The framework represents what the SSM parameterisation supports without claiming spatial flow dynamics the monitoring data cannot constrain.

S3.4 Climate Scenarios in the Per-Well Framework

Climate scenarios applied under the per-well framework are based on the UKCP18 Regional 12 km projections for Wales (Met Office, 2018) and the CHESS-SCAPE bias-corrected projections of Robinson et al. (2023). Two scenarios are carried forward into the Results (Section 4.13): UKCP18 2050s (2040–2069) applies a winter precipitation increase of +10%, summer precipitation decrease of −15%, winter PET increase of +5% and summer PET increase of +20%; UKCP18 2080s (2070–2099) applies +20%, −30%, +10% and +35%

respectively for the same four terms. Both scenarios use RCP8.5 central-estimate (50th-percentile) values. Perturbations are applied per season, with winter defined as November–March and summer as May–September; the climatological baselines against which the perturbations are applied are the 2005–2026 monthly means from the RAF Valley climate record. The resulting Δh for each well is computed for each season separately, and the annual Δh is reported as the 0.5-weighted mean of the winter and summer equilibrium responses — this convention captures the physically significant seasonal asymmetry between winter recharge and summer evaporative loss that an annual-mean forcing would obscure.

Three limitations on the interpretation of these outputs apply. First, the perturbations used are central-estimate values; the UKCP18 probabilistic range at the 10th and 90th percentiles spans roughly half to twice the central perturbation for precipitation and a broader range still for PET, so the Δh values reported here are a central point within a substantially wider envelope of plausible responses. Users of the interactive scenario viewer (scenario_viewer.html) can explore perturbations across the full 0.5–1.5× range of the seasonal sliders to visualise this envelope. Second, the equilibrium framework computes the steady-state response to sustained seasonal perturbations and does not resolve within-year dynamical propagation of recharge between winter and summer — for example, the effect of a wetter winter on the summer minimum through carry-over storage is captured only to the extent that the SSM β_3 drainage coefficient encodes it in the fitted mean. Third, the framework holds the SSM β coefficients themselves fixed at their historical values; it does not allow for structural changes in the β coefficients under future climate (for example, a higher β_2 under consistently warmer summers through increased vapour pressure deficit). Within these limitations, the climate scenario output is the most quantitatively defensible use of the equilibrium framework: every cluster's response is derived from its own fitted β coefficients under a uniform climate perturbation that reflects the current best estimate for the site's regional climate trajectory, and the result reflects the physical partition of rainfall and PET sensitivity that the SSM has estimated from 21 years of observation.

S3.5 Forest Management Scenarios in the Per-Well Framework

Forest management scenarios (full clearfell, 50% thinning, broadleaf conversion) apply canopy parameter changes only to the C4 and C5 Forest wells, and the framework correctly produces a response confined to the C4/C5 zone. This is a feature of the framework, not a bug: under the SSM parameterisation, canopy management directly affects only the wells beneath the canopy, and there is no structural mechanism for canopy effects to propagate into the open dune clusters beyond the hydraulic gradient already implicit in the baseline heads. The BACI monitoring record provides empirical corroboration of this structural feature: post-felling displacement is concentrated at the core impact wells, and wells outside the C4 cluster show no detectable felling-specific response once common-mode climate variability is accounted for (§4.6.4).

The clearfell scenario applies a β_2 multiplier of ×0.952 at C4 and C5 wells: the felled well's own post-felling β_2 ratio net of the climate controls (Impact ratio 0.862 minus Climate Control drift 0.910, plus 1.0; Script 10e). Canopy interception is unchanged, because the felled well's recharge did not rise (D-239). The response is reported as the sustained change in level: +0.038 m (Model B) and +0.066 m (Model A) at C5, +0.217 m at C4 (Model A), and +0.060 m and +0.135 m at the felled well itself, against the +0.108 m measured there. This is the lower end of the clearfell range. The 4.2 ha compartment is narrower than the drawdown reach, so a recharge gain there would spread into the surrounding forest: at WMC3 a disc of its size keeps about 15% of a full local gain (Script 20), which the felled well's β_1 cannot distinguish from none. Returned in full, as for a felling wider than the reach, the intercepted rain gives the upper end: +0.435 m (Model B) and +0.986 m (Model A) at C5, and +1.891 m

at C4 (Model A). Under Model A these exceed the open-dune end state (+0.490 m at C5, +0.777 m at C4), which bounds what removing the canopy can do.

The thinning scenario applies half the clearfell perturbation ($\beta_2 \times 0.976$), again with interception unchanged, giving half the clearfell response. That is an upper bound: after thinning the remaining trees and the understorey take up part of the water released, and the saving shrinks as the stand recovers (Gebhardt et al., 2014).

The broadleaf conversion scenario applies the seasonal β_2 profile (winter ×0.882, summer ×1.075) with interception unchanged (D-239). Weighted by evaporative demand, the summer increase outweighs the winter saving, and the framework predicts a small fall in level at both forest clusters (−0.037 m at C5 and −0.121 m at C4 under Model A). With the lower interception of a broadleaf canopy (15%, following the temperate deciduous meta-analysis of Komatsu et al., 2011) returned as recharge, it predicts a rise instead (+0.308 m at C5 and +0.506 m at C4 under Model A); catchment modelling on sandy soils favours that end (Sonnenborg et al., 2017). The phenological mechanism of §5.5.2 — a larger winter recharge pulse under leafless canopy, drained before summer — is a dynamical trajectory response that the equilibrium framework does not resolve.

For the Figure 82 hydrograph (§5.7.4), a seasonally-varying β_2 is applied to the broadleaf scenario to represent the full-LAI summer canopy and leafless winter canopy. This seasonal parameterisation is a visualisation device and is not part of the §4.13 Results; it is used specifically to communicate the phenological argument from §5.5.2 alongside the BACI-observed record. The annual-mean output in 19_scenario_summary.csv remains the quantitatively citeable framework result.

S3.6 Implications and Future Work

The per-well equilibrium framework is a lightweight analytical device for translating the SSM β coefficients into spatial summaries of equilibrium head response. Its strengths are that it makes no claims beyond what the SSM itself supports, that it is fully reproducible from the published CSV outputs, and that it is explorable through an interactive viewer that exposes all parameter choices to the user. Its limitations are those of any equilibrium framework: it does not resolve dynamical trajectory responses, it does not capture cluster-to-cluster hydraulic coupling, and it cannot simulate transient interventions in which the time evolution of the water table matters for ecological outcome. The framework is adequate for the questions the paper addresses — whether forest management in the C4/C5 zone affects summer minima in C1 and C2, and how climate perturbations partition across the cluster structure — and the BACI monitoring record confirms the framework's structural conclusions.

Answering a different class of question, such as how a hypothetical lateral flux along the CEH14 ridge margin would propagate through the aquifer, would require a calibrated continuous-flow model. For that, the critical unknowns are hydraulic conductivity (currently constrained by a single tracer test at K = 6.0 m/day; Betson et al., 2002) and aquifer thickness (inferred from four borehole logs across 700 ha). Slug tests at two to three representative wells per cluster and a ground-penetrating radar survey of aquifer thickness are identified as the priority field measurements required to enable a properly calibrated continuous-flow model, should future work pursue intervention scenarios where lateral hydraulic propagation is central to the management question.

Summary of scenario modelling approach
Primary framework: Per-well equilibrium with triangulation-linear grid rendering (Sections 3.8, 4.10; scenario_viewer.html; 19_scenario_summary.csv)

Evaluated and rejected: 2D steady-state Helmholtz PDE solver (Section 3.8 PDE paragraph; S3.3 above)
Visualisation device: Figure 82 synthetic hydrograph with seasonally-varying broadleaf β_2 (§5.7.4; Script 21)
Empirical corroboration: BACI monitoring record (§4.6)

S3.7 Interactive Viewer: Rendering Conventions and Depth-Mode Visualisation

The companion interactive scenario viewer (scenario_viewer.html; Hollingham, 2026) renders the per-well Δh field described in S3.2, but uses a distinct grid-rendering method at the browser to preserve responsiveness under slider-driven re-interpolation. The viewer implements inverse-distance weighting with power 1 and eight nearest neighbours (10 m distance floor; 5 m exact-value shortcut) rather than the triangulation-linear method used for the static figures. Power-1 IDW with a fixed neighbour set produces a smoother surface than the naïve power-2 IDW previously used, removes the bullseye halo artefacts that arose around individual wells on the Δh surface, and is sufficiently close to the triangulation-linear rendering at well locations that the two methods agree to within the mapped colour-scale resolution across the interpolated interior. At well locations the viewer returns the well's observed value unchanged; outside the convex hull of the wells it extrapolates under the IDW kernel where the triangulation-linear method would return no data. This distinction is most visible near the south-eastern coastal margin, where the viewer renders a short extrapolated zone that the static figures correctly leave blank.

The viewer offers three complementary display modes. The first two — change in water table elevation (Δh) and absolute water table elevation (m AOD) — are elevation-based surfaces, and are the two quantities exported to 19_scenario_summary.csv. The third mode, depth below surface (m), is a viewer-specific visualisation derived in the browser. It is computed per grid cell as $d_w = z_{DEM}(E, N) - z_{WT}(E, N)$, where z_{DEM} is the bilinearly-interpolated LiDAR DEM elevation at cell centre and z_{WT} is the IDW-interpolated scenario water table elevation at the same cell. This per-cell subtraction is the quantity that mediates the ecological response: the Curreli et al. (2013) slack vegetation thresholds are defined in depth-below-surface terms (SD15b wet slack summer viability limit: 0.61 m; SD16 dry slack viability limit: 0.98 m), and rendering the scenario water table against the topographic surface permits direct visual identification of the slack positions where scenarios cross these thresholds. The viewer's depth-mode colour scale is anchored on these thresholds: light blue at the 0.10 m winter flooding limit, green at 0.61 m, orange at 0.98 m, and deep red at 1.50 m, with explicit black tick marks on the legend at the SD15b and SD16 threshold depths.

A dune-ridge mask is applied exclusively in depth-below-surface mode. The rationale is that the water table is a continuous surface in space, and its elevation relative to ordnance datum is well-defined at every point within the site regardless of whether the overlying ground is a ridge or a slack floor; elevation-based maps (modes one and two) therefore require no topographic masking. Depth below surface, however, is meaningful only where the ground is approximately at the well-plane elevation — at a dune ridge the interpolated water table sits several metres below ground, but this is a consequence of the ridge height, not of any hydrological feature, and rendering such cells would produce a misleading strong-red patch on the depth map that obscures the ecologically relevant slack-floor response. The viewer masks a cell in depth mode when the DEM elevation at that cell exceeds the IDW-interpolated well-plane elevation (computed by the same IDW kernel applied to well DEM ground elevations) by more than 1.0 m. This threshold is user-toggleable from the viewer's control panel; disabling it reveals the unmasked depth field, which can be useful for diagnostic inspection but should not be used for ecological interpretation.

Additional viewer conventions of note: the basemap is a LiDAR-derived greyscale hillshade rendered at approximately 35% opacity underneath the scenario surface, matching the static figure hillshade convention of map_utils.load_dem_hillshade with azimuth 315°, altitude 35° and vertical exaggeration 3; climate perturbations can be explored across the full 0.5–1.5× range per season, extending substantially beyond the UKCP18 2050s and 2080s central-estimate presets to allow sensitivity analysis across the probabilistic envelope; and tooltips on any well display both the baseline and scenario-perturbed depth values alongside the absolute head and Δh, providing a direct visual link between the per-well framework output and the ecological depth-below-surface interpretation.

S3.8 Reproducibility

The per-well equilibrium framework and climate/forestry scenario calculations are implemented in 19_spatial_groundwater.py, which exports the full per-scenario × per-season × per-cluster output to 19_scenario_summary.csv and generates the interactive scenario viewer. The interactive scenario viewer (scenario_viewer.html) implements the browser-side IDW rendering described in S3.7 and reads the same per-well data. The synthetic mean-year hydrograph (Figure 82) is generated by 21_forestry_scenarios.py, which reads BACI coefficients from the Script 10 clearfell suite outputs. All scenario parameters — UKCP18 perturbation fractions, canopy interception values, β_2 multipliers, and seasonal definitions — are defined as named constants at the head of each script and are fully documented in the scenario_viewer.html source.

Supplementary Note S4: Specific Yield Mapping

S4.1 Per-Well Estimates

Table S4.1. Individual Well WTF Specific Yield Estimates — Newborough Warren 2005–2026: event-based median specific yield (Sy) derived from the water table fluctuation (WTF) method (Healy and Cook, 2002) for the 66 reference-network wells. Cl. = cluster under the k = 5 partition (C1 Lake Edge, C2 Dune, C3 Western Residual, C4 Main Forest, C5 Coastal Forest). n = number of qualifying monthly rising-limb events (criteria: Δh > 5 mm, net recharge P − PET > 10 mm). Q25/Q75 = interquartile range; a wide IQR reflects month-to-month variability in event estimates rather than measurement error. Wells in the forest clusters C4 and C5, marked Int. corr. = Yes, have net recharge adjusted for 24% canopy interception (Freeman, 2008). CEH12 (bedrock ridge — WTF response reflects fractured rock) and CEH15 (forest slack floor — slack topography dominates water table dynamics) are excluded from the IDW interpolation surface and do not appear here. Source: 18_wtf_01_well_sy_estimates.csv.

Well	Cl.	n	Sy med	Sy Q25	Sy Q75	Int. corr.
CEH11	C1	56	0.236	0.164	0.311	
CEH23	C1	42	0.206	0.111	0.252	
CEH25	C1	55	0.179	0.132	0.267	
CEH26	C1	48	0.212	0.114	0.271	
CEH27	C1	55	0.225	0.144	0.315	
CEH5	C1	66	0.201	0.159	0.277	
CEH6	C1	68	0.211	0.122	0.345	
D10	C2	49	0.277	0.224	0.377	
D17	C2	55	0.273	0.204	0.362	
D41	C2	46	0.248	0.204	0.317	
D43	C2	61	0.295	0.212	0.347	
D44	C2	50	0.228	0.163	0.319	
D5	C2	50	0.279	0.215	0.348	
D6	C2	58	0.268	0.200	0.365	
D7	C2	52	0.290	0.237	0.331	
D8	C2	47	0.262	0.200	0.381	
L7	C2	64	0.292	0.241	0.357	
T41B	C2	46	0.233	0.174	0.322	
T41C	C2	47	0.229	0.200	0.316	
T41D	C2	47	0.268	0.204	0.339	
WMC1	C2	51	0.283	0.227	0.417	
CEH24	C2	52	0.252	0.179	0.327	
CEH28	C2	41	0.244	0.189	0.313	
NW3	C2	52	0.278	0.199	0.355	
NW4	C2	55	0.255	0.167	0.337	
NW4B	C2	56	0.252	0.161	0.328	
D15	C3	54	0.290	0.237	0.363	
D25	C3	46	0.312	0.262	0.396	
D38	C3	53	0.222	0.180	0.288	
D9	C3	50	0.273	0.207	0.373	
T41A	C3	55	0.245	0.178	0.336	
WMC2	C3	52	0.352	0.306	0.419	
CEH1	C3	69	0.274	0.225	0.347	
CEH10	C3	63	0.264	0.164	0.372	
CEH18	C3	57	0.366	0.288	0.439	
CEH21	C3	27	0.400	0.353	0.440	
CEH36	C3	36	0.366	0.311	0.428	
CEH39	C3	37	0.350	0.302	0.403	

Well	Cl.	n	Sy med	Sy Q25	Sy Q75	Int. corr.
CEH4	C3	52	0.350	0.252	0.400	
CEH40	C3	28	0.286	0.247	0.320	
CEH41	C3	38	0.307	0.261	0.355	
CEH42	C3	35	0.352	0.290	0.386	
CEH9	C3	52	0.340	0.272	0.389	
NW1	C3	73	0.258	0.210	0.355	
NW11	C3	69	0.284	0.226	0.356	
NW13	C3	48	0.253	0.214	0.344	
NW2	C3	69	0.254	0.212	0.357	
NW5	C3	57	0.311	0.250	0.350	
NW6	C3	47	0.282	0.218	0.365	
NW7	C3	48	0.333	0.284	0.404	
WMC3	C3	38	0.339	0.289	0.427	
WMC4	C3	52	0.257	0.210	0.349	
CEH32	C4	50	0.242	0.181	0.355	Yes
CEH13	C4	51	0.231	0.175	0.317	Yes
CEH14	C4	56	0.199	0.128	0.265	Yes
CEH2	C4	60	0.236	0.192	0.308	Yes
CEH20	C4	52	0.253	0.217	0.325	Yes
CEH30	C4	50	0.274	0.215	0.350	Yes
CEH33	C4	50	0.259	0.204	0.348	Yes
CEH34	C4	51	0.261	0.211	0.340	Yes
NW10	C4	67	0.262	0.195	0.341	Yes
CEH16	C5	52	0.285	0.241	0.395	Yes
CEH17	C5	40	0.322	0.246	0.400	Yes
CEH19	C5	41	0.341	0.278	0.426	Yes
CEH31	C5	43	0.294	0.234	0.384	Yes
NW9	C5	62	0.308	0.243	0.403	Yes

S4.2 Reproducibility

The per-well WTF specific yield estimates tabulated in Table S4.1 are computed by 18_wtf_spatial.py (wtf_individual_wells), which identifies qualifying monthly rising-limb events, applies the canopy interception correction to the C4 and C5 forest wells, and exports the per-well statistics to 18_wtf_01_well_sy_estimates.csv — the file the table's caption names. The same script performs the spatial interpolation of Sy and the exclusion of CEH12 and CEH15 from the interpolated surface. 17_wtf_specific_yield.py is the cluster-level estimator and exports 17_wtf_01_sy_estimates.csv; it does not produce the per-well table. Its earlier output name, 17_wtf_well_sy.csv, was retired under D-038 on 2026-08-19; the two files were byte-identical while both existed, and the name was superseded rather than the content changed. 18_wtf_spatial.py reads 01_wells_clean.csv, 01_climate.csv, 01_locations.csv and 02_cluster_stats.csv; both scripts are maintained in the canonical pipeline sequence.

Supplementary Note S5: Residual-Lag Test for Ridge-Derived Recharge

S5.1 Purpose

An earlier boundary-subsidy argument attributed a persistent water-balance residual at forest-margin wells — most visibly a positive residual at CEH14 — to lateral recharge derived from the northern rock ridge. It rested on two lines of evidence: a spatially structured pattern of positive residuals concentrated along the forest–dune boundary, and a closure argument that no other plausible flux could account for the monthly deficit. The corrected residual field (Script 20, 2026-08-06; §4.9.6, Figure 59) removes both. Sixty-five of the sixty-six reference wells now fall within ±0.01 m/month, the field shows no gradient on either axis, no well exceeds +0.02 m/month, and CEH14 is the most negative well in the network at −0.010 m/month. The balance closes without requiring an additional flux. The residual-lag analysis described below was designed as an independent, falsifiable test of the transport mechanism and is retained as a bound on what the 21-year monthly record could have detected: if ridge-derived recharge is delivered as a time-varying pulse, then water travelling from the ridge to successive dune wells must arrive at those wells with travel times that increase as a function of distance. The time structure of the SSM residuals, cross-correlated against rainfall, should therefore carry a distance-dependent lag signature. This Note documents the test and reports the null result.

S5.2 Extended State-Space Model

The single-period SSM fitted in Section 3.4.3 expresses the water table increment as a function of contemporaneous rainfall, where h_disp_prev(t) = DRAINAGE_DATUM + h(t−1) is the displacement above the 3.7 m drainage datum at the start of month t:

$$\Delta h(t) = \alpha_B + \beta_1 \cdot P(t) - \beta_2 \cdot PET(t) - \beta_3 \cdot h_disp_prev(t) + \varepsilon(t)$$

A preliminary analysis (Script 22) demonstrated that the residuals ε(t) from this formulation carry a generic lag-1 rainfall signal at every well on the site, irrespective of cluster or position. This is the expected monthly-timestep consequence of the vadose zone: recharge takes approximately one month to propagate to the water table, so monthly-averaged water-table response lags monthly-averaged rainfall by roughly one month. This generic signal dominates the cross-correlation function at every well and masks any ridge-specific structure that might be present at longer lags.

To remove this confound, the residual-lag test uses an extended model that explicitly includes both P(t) and P(t−1) as regressors:

$$\Delta h(t) = \alpha_B + \beta_{10} \cdot P(t) + \beta_{11} \cdot P(t-1) - \beta_2 \cdot PET(t) - \beta_3 \cdot h_disp_prev(t) + \varepsilon'(t)$$

The β_{10} and β_{11} coefficients absorb the vadose-zone response between them; any remaining lag structure in ε′(t) is therefore post-vadose and is the candidate ridge-transport signal. Fitted coefficients show that the extended parameterisation is only partly identified: β_{11} is statistically significant ($p < 0.05$) at 32 of the 66 wells, and its sign is mixed (negative at 33 wells, positive at 33), so the two rainfall terms do not partition the vadose-zone response cleanly at every well. The cluster-median lagged-response fraction $\beta_{11} / (\beta_{10} + \beta_{11})$ nonetheless rises across the open-dune-to-forest sequence, from −0.08 in C1 and −0.03 in C2 to +0.08 in C3 and +0.27 in C4 (C5 +0.16), consistent with the slower drainage of forest soils. Mean R^2 rises only marginally, from 0.758 under the single-period model to 0.768 under the extended model across the 63 wells fitted by both: the second rainfall term buys very little explanatory power, which is consistent with β_{10} and β_{11} dividing one vadose-zone response between them rather than capturing two distinct processes. Per-well fits are in 23_ridge_lag_fits.csv and 22_model_b_fits.csv. The extended model is used solely for this

diagnostic test; the β_1, β_2, β_3 and α_B values reported in the main text remain the authoritative parameterisation for all other analyses.

S5.3 Cross-Correlation and Pre-Whitening

For each well with at least 140 months of data (n = 63 after excluding CEH3, CEH4, CEH7, CEH8, CEH37 and llynrhos for reasons noted below), the Pearson cross-correlation r(ε′(t), P(t − N)) was computed for lags N = 0 to 12 months. Prior to cross-correlation, both the rainfall and residual series were pre-whitened using a first-order autoregressive filter: x′(t) = x(t) − φ·x(t−1), where φ is the AR(1) coefficient of the rainfall series (+0.207 at RAF Valley). Wells whose residuals retained AR(1) structure above |φ| = 0.2 were additionally pre-whitened against their own residual autocorrelation before the common filter was applied. Statistical significance was evaluated using the Bartlett 95% confidence interval for a white-noise series, |r| ≥ 1.96/√N, which for typical series lengths gives a threshold of approximately 0.15.

Wells excluded from the test: CEH3 (tidal boundary — outside the SSM operational domain), CEH4 (coastal erosion drift plus post-2017 clearfell drawdown confounding any lag signal), CEH7, CEH8, CEH37 (standard upstream exclusions carried over from §4.10.5); llynrhos (lake-stage well, not a dune-aquifer observation). Wells beyond the 140-month minimum were retained without further filtering.

S5.4 Hypothesis Test

The test metric is the Spearman rank correlation between each well's peak-correlation lag N* and its Euclidean distance from the ridge reference point at (E = 241750, N = 364500, OSGB36) — a representative coordinate for the nearest point of the northern rock ridge to the dune field. Under the ridge-transport hypothesis, Spearman ρ should be significantly positive.

Of 66 wells fitted, 49 had a peak cross-correlation that exceeded the Bartlett significance threshold. Across these 49 wells the peak lag has a mean of 2.49 months, a median of 2 months and a standard deviation of 1.91 months: 42 wells peak at lag 2, five at lag 3, one at lag 11 and one at lag 12, with no systematic geographic pattern. The Spearman rank correlation between peak lag and ridge distance was ρ = −0.028 (p = 0.85) — no distance structure whatever. Cluster mean peak lags are 2.00 months at C1, C2 and C3 and 3.67 at C4; one C5 well reaches the significance threshold, at lag 11. Forty-eight of the forty-nine carry a cluster assignment — CEH22 is fitted and significant but sits outside the k = 5 partition, so the per-cluster counts sum to 48. A Mann-Whitney comparison of the C4 ridge-adjacent wells against the C2 and C3 dune-body wells does return significantly longer lags at C4 (3.67 against 2.00 months; U = 232.5, p < 0.001, computed from the peak lags in 23_ridge_lag_fits.csv), but that ordering is a cluster-level contrast in drainage behaviour rather than a distance effect: within the network as a whole, lag is uncorrelated with ridge distance. The hypothesis of distance-dependent ridge transport is not supported.

The magnitude of coupling does not single out the ridge margin. Mean peak |r| among significant wells is highest at C1 (0.269) and C2 (0.222), and the ridge-adjacent C4 wells (0.190) sit within the same narrow band as the geologically separate C3 wells (0.182). At the individual wells previously highlighted, CEH14 peaks at r = +0.21, CEH34 at +0.20, CEH2 at +0.18 and CEH13 at +0.16, against NW10 at +0.19 and NW1 at +0.13 in C3 — a spread too narrow to support a ridge-specific coupling signal. Neither the strength of rainfall coupling nor its timing is distance-structured.

S5.5 Interpretation

The null distance-lag result does not rule out ridge-derived recharge. It rules out one specific form of it: event-driven, pulse-delivered transport in which monthly rainfall on the ridge propagates to the dune field with travel times that scale with distance. The alternative interpretation — that ridge recharge is delivered as a near-steady baseflow, sufficiently smoothed by its transit through fractured bedrock that its month-to-month variance falls below the detection threshold of monthly water-level records — is consistent with the null. A steady baseflow of this kind would manifest observationally as a constant positive contribution to the local water balance at wells along the flow path, which is precisely what the SSM residual α_B already absorbs. That predicted signature is, however, absent from the corrected residual field: the C4 median is −0.007 m/month and no forest well carries a residual above +0.02 m/month. The steady-baseflow reading is therefore consistent with the lag-test null but unsupported by the water balance, and the two mechanisms remain indistinguishable from water-level data alone only in the weak sense that neither is evidenced.

The main report's treatment of the water-balance residual (§5.2.1) no longer advances a ridge contribution: with the corrected field the balance closes without requiring an additional flux, and the residual is discussed there as a model-adequacy diagnostic rather than as evidence of a boundary subsidy. Definitive resolution of the transport mechanism would require either a ridge-crest rain gauge paired with bedrock piezometers, or direct geochemical tracer work on the ridge–dune flow path. Both are identified as priority further work in §5.9.

S5.6 Reproducibility

The residual-lag test is implemented in two standalone analysis scripts, 22_residual_lag_analysis.py and 23_ridge_recharge_lag_test.py, which are maintained as supplementary diagnostics (Phase 12) alongside the main pipeline to preserve the reproducibility of the headline results from the canonical pipeline sequence. Script 22 generates the residuals and AR(1) diagnostics; Script 23 fits the extended model, computes the cross-correlations, and applies the Spearman test. Both scripts read the same intermediate data produced by the main pipeline (01_wells_clean.csv, 01_climate.csv, 01_locations.csv, 02_cluster_stats.csv) and produce independent outputs under 22_residual_lag_analysis/ and 23_ridge_recharge_lag_test/ in the project outputs tree. A plain-text summary of the hypothesis test result is written to 23_05_hypothesis_test_summary.txt at runtime.

Supplementary Note S6: Seasonal Climatology Diagnostic of the Residual

S6.1 Purpose

Supplementary Note S5 reported a null result for event-driven ridge transport. This Note applies the same logic to a second candidate mechanism that was advanced when the water-balance residual was believed to carry spatial structure: that the Thornthwaite PET estimate underestimates the true summer atmospheric demand on the water table because it is temperature-only and does not capture net radiation, vapour pressure deficit, wind, or surface-condition effects. As with S5, the corrected residual field leaves no anomaly for this mechanism to explain, and the diagnostic is retained for what it establishes about the monthly time-series residual ε(t) rather than as a competing explanation. Thornthwaite PET is an empirical index of atmospheric evaporative demand derived from temperature and day length; it does not define a specific reference surface. If that were the dominant explanation, the residuals should carry a systematic summer-negative signature that the fitted β_2 has not absorbed. This Note tests for that signature, along with two subsidiary diagnostics that bear on the remaining candidate explanations (steady ridge baseflow, nonlinear recharge response, and residual model error). The residual quantity throughout S5 and S6 is the monthly time-series SSM residual ε(t), which is distinct from Script 20's scalar per-well water-balance closure residual discussed above and is unaffected by the 2026-08-06 correction to the latter; only the framing of these two Notes changes.

S6.2 Diagnostics

Three complementary tests are computed from the residuals ε(t) produced by Script 22 for the same 63 wells that passed the ≥ 140-month record-length filter and the site-extent exclusions detailed in S5.3.

(i) Seasonal climatology. For each well, the monthly climatology of the residual is computed as the mean of ε(t) by calendar month over the record, giving a 12-point annual cycle. A sinusoidal fit $\varepsilon(m) = a_0 + a_1 \cdot \cos(2\pi m/12) + a_2 \cdot \sin(2\pi m/12)$ yields an amplitude $\sqrt{(a_1^2 + a_2^2)}$ and a phase (month of peak). Summer-minus-winter magnitude (JJA mean − DJF mean) is reported as a robust non-parametric proxy for the same quantity.

(ii) Independent ET proxy. A correlation of the residual against PET itself would be trivially zero because OLS fitting makes residuals orthogonal to every regressor by construction. Instead, the residual is correlated against monthly sunshine hours from the RAF Valley record — a direct radiation-based measurement that is not in the regression. If Thornthwaite underestimates summer ET in a way that β_2 has not absorbed, high-insolation months should carry disproportionately negative residuals, yielding a systematic negative Pearson r.

(iii) Spatial pattern within clusters. For C3 (Western-block open dune), the cluster is split by Euclidean distance from the ridge reference point used in S5 at a 1 km threshold (7 forest-adjacent wells, 10 warren-interior wells), and a Mann-Whitney test asks whether forest-adjacent wells have systematically smaller seasonal amplitudes than warren-interior wells — as would be expected if a steady ridge baseflow were flattening the residual at ridge-proximal locations.

S6.3 Result: the ET Hypothesis Is Not Supported

The sunshine-hours correlation is consistently negative in sign across every cluster, with network mean r = −0.035 and per-cluster means ranging from −0.010 (C4 Main Forest) to

−0.056 (C1 Lake Edge). None of the 63 wells exceeds the Bartlett 95% significance threshold of |r| = 0.15 — all 63 fall inside the null band. The sign is directionally consistent with Thornthwaite slightly underestimating radiation-driven ET, but the magnitude of the bias is below the resolution of the monthly water-level data to detect.

Summer-minus-winter residuals are negative in every cluster, with magnitudes ranging from −0.001 m at C5 to −0.010 m at C3 (C1 −0.005 m, C2 −0.004 m, C4 −0.008 m). These values are an order of magnitude smaller than the month-to-month variability of the residuals themselves (typical well-level residual standard deviation of order 0.1 m) and should not be read as a summer-ET signal. Notably, C4 — the cluster where canopy interception would most plausibly produce Thornthwaite miscalibration — has the weakest sunshine correlation of any cluster, indicating that the cluster-mean β_2 has absorbed forest-specific ET behaviour reasonably well despite the simplicity of the fitted model.

The C3 within-cluster split also returns a null: forest-adjacent C3 wells (n = 8) have smaller seasonal amplitudes than warren-interior wells (n = 14) — 0.0109 m against 0.0118 m; Mann-Whitney U = 50.0, p = 0.36, alternative “adjacent < interior”. The direction is the one a steady-ridge-baseflow mechanism would predict, but the difference is under a millimetre and not significant, so it carries no evidential weight either way.

S6.4 What the Residuals Actually Look Like

The per-cluster seasonal climatologies reveal a structure that fits none of the candidate mechanisms cleanly: the annual cycle is bimodal, with positive residuals in January–February AND June–July, and negative troughs in April–May and September–October. The sinusoidal fit captures this poorly, which is why the fitted amplitudes appear modest relative to the visible annual range. The fitted sinusoid peaks in late autumn or winter in every open-dune cluster — C1, C2 and C3 all in December, C5 in November — and later, in February, at C4, the shift at the forest cluster being consistent with its slower drainage. Across the network, 50 of the 63 wells peak between November and March and not one peaks in the May–August summer band.

This bimodal structure is not the signature of unmodelled summer ET, which would produce a single summer trough. It is not the signature of flat steady ridge baseflow, which would produce no annual cycle at all. And it is not a simple winter-recharge-nonlinearity signature, which would produce a single winter peak. Its most parsimonious interpretation is that a linear lumped-parameter model with time-invariant β coefficients is a rough approximation of a system whose soil-moisture storage operates nonlinearly across the annual cycle — wet-winter saturation increasing recharge efficiency beyond what the cluster-mean β_1 represents, shoulder-season soil drying reducing it below, and the whole cycle modulated by seasonal variation in vegetation water use that a constant β_2 cannot track.

The spatial pattern of seasonal amplitude nonetheless shows a coherent structure: the two largest amplitudes in the network are CEH14 (0.040 m) and CEH13 (0.026 m), both C4 forest-margin wells, against a network median of order 0.010 m; C4 has the largest cluster-mean amplitude (0.015 m) of any cluster. The two fields do not coincide, however: CEH14 carries the most negative water-balance residual in the network (−0.010 m/month; Figure 59, §4.9.6), at the opposite end of that distribution from the open-dune wells that lead it (NW3, CEH23, NW6). The rest of the C4 cluster sits within the general network spread rather than standing out. The elevated seasonal variance at CEH14 and CEH13 is therefore not accompanied by an elevated water-balance residual, which weakens the reading that larger seasonal boundary fluxes are responsible and leaves the alternative — that cluster-mean β values are a worse approximation at ridge-margin wells than at dune-interior wells — as the better-supported of the two. Water-level data alone cannot settle it.

S6.5 Combined Conclusion from S5 and S6

Taken together, Supplementary Notes S5 and S6 have excluded the two leading specific attributions of the water-balance residual:

The residual does not show the distance-dependent lag signature of event-driven ridge transport (S5).
The residual does not show the summer-negative signature of unmodelled Thornthwaite underestimation of atmospheric demand (S6).

The main text of §5.2.1 has been framed accordingly: the model explains 50–70% of month-to-month variance in Δh depending on cluster, and the unexplained portion is real and spatially structured (concentrated at the forest margin) but cannot be uniquely attributed to any single physical mechanism from water-level data alone. The fitted residual α_B remains the most defensible summary statistic for the steady-state boundary contribution; the underlying mechanism is most plausibly a combination of steady ridge baseflow, nonlinear soil-moisture storage dynamics not captured by the linear SSM, and minor β coefficient miscalibration, in proportions that cannot be resolved without additional measurement of the kind identified as priority further work in §5.9.

S6.6 Reproducibility

This analysis is implemented in 24_residual_seasonality.py, which like Scripts 22 and 23 is maintained as a supplementary diagnostic (Phase 12) alongside the main pipeline to preserve the reproducibility of the headline results. The script reads the same intermediate data as Script 22 (01_wells_clean.csv, 01_climate.csv, 01_locations.csv, 02_cluster_stats.csv) plus the raw RAF Valley climate file for sunshine hours (RAF_Valley_Climate.csv). All outputs are written to 24_residual_seasonality/ in the project outputs tree: a per-well climatology table (24_residual_climatology.csv), the per-cluster climatology panels (24_01_climatology_panels_by_cluster.png), the spatial amplitude map (24_02_seasonal_amplitude_map.png), the sunshine-correlation scatter (24_03_sun_residual_correlation.png), the per-cluster phase distribution (24_04_phase_by_cluster.png), and a plain-text summary of the diagnostic result (24_05_diagnostic_summary.txt).

Supplementary Note S7: Equilibrium Wetness Index — Per-Well Reconstruction

S7.1 Per-Well Reconstruction

The equilibrium wetness index (EWI) is the steady-state water-table level implied by each well's fitted state-space coefficients under long-term mean climate. Because its forcing is the long-term normal rather than a specific observation window, it is climate-window-independent and computable from a shorter record than the five-year mean spring level (MSL5) requires. Section 3.7.6 of the main report defines the index and its uncertainty; Section 4.8.4 reports its calibration onto the MSL5 scale and its cross-validation against the Ellenberg-F vegetation data. This note carries the full per-well reconstruction that Section 4.8.4 refers to.

The calibration is scoped to the open-dune network (clusters C1–C3, reference and extended wells). Of the 63 open-dune wells carrying an index, 61 also carry an observed MSL5 and set the calibration; the remaining two have no valid five-year spring window and are reconstructed only. The 21 forest wells (C4, C5) are reconstructed but held out of scope: their coefficients are the least constrained on the site, and the reconstruction there is too coarse to place a well across a Curreli et al. (2013) slack threshold. All 84 wells are listed, with the scoping carried explicitly in the Status column, so that the boundary of the calibration is visible rather than implied. CEH13 and CEH14 do not appear: the drainage coefficient is the denominator of the equilibrium expression and is near zero or negative at both, leaving the index undefined.

Standard errors propagate uncertainty in the fitted coefficients through the equilibrium expression. The variant tabulated here propagates the drainage coefficient alone, which is the dominant term because the equilibrium displacement is inversely proportional to it; a first-order propagation over all three coefficients is carried alongside it in the source CSV. The propagation is anchored on the equilibrium displacement rather than on the datum-referenced index value: subtracting the constant drainage datum shifts the value but not its uncertainty, and anchoring on the shifted value understates the error, severely so at wells whose equilibrium level sits close to the datum. These standard errors are substantially larger than the sampling error on a five-year MSL5 mean at every well, which is the basis for treating the index as a complement to MSL5 rather than a replacement for it.

Table S7.1. Equilibrium wetness index and MSL5 reconstruction, per well (n = 84). Levels are metres below ground surface, negative below the surface. The index is reconstructed from the open-dune calibration MSL5 = +0.392 + 0.955·EWI (r = 0.96, RMSE = 76 mm), fitted on the 61 in-scope wells carrying an observed five-year mean spring level. A further 2 open-dune well(s) carry an index but no valid five-year spring window and are reconstructed only; the 21 C4/C5 forest wells are reconstructed but held out of scope, their coefficients being the least constrained on the site. Status reads: 'Calibration', an in-scope well that entered the fit; 'Reconstructed', in scope but with no observed spring level; 'Out of scope', a forest well outside the calibration's scope. Standard errors on the index propagate the drainage coefficient alone, the dominant term; the full three-coefficient variant is carried in 26_equilibrium_wetness_index_per_well.csv. Source: 26_equilibrium_wetness_index_per_well.csv, 26_ewi_msl5_comparison.csv.

Well	Network	Cluster	β_3 (mth^{-1})	EWI (m bg)	SE (mm)	MSL5 obs	MSL5 recon	Resid. (mm)	Status
CEH11	Reference	C1 (Lake Edge)	0.0925	-0.636	312	-0.233	-0.219	15	Calibration
CEH23	Reference	C1 (Lake Edge)	0.0940	-0.549	332	-0.159	-0.133	26	Calibration
CEH25	Reference	C1 (Lake Edge)	0.1343	-0.511	296	-0.107	-0.097	10	Calibration
CEH26	Reference	C1 (Lake Edge)	0.0876	-0.513	346	-0.149	-0.099	50	Calibration
CEH27	Reference	C1 (Lake Edge)	0.0984	-0.439	303	-0.045	-0.027	18	Calibration
CEH5	Reference	C1 (Lake Edge)	0.0983	-0.530	300	-0.114	-0.115	-1	Calibration
CEH6	Reference	C1 (Lake Edge)	0.1222	-0.407	263	-0.049	0.005	54	Calibration
CEH12	Extended	C1 (Lake Edge)	0.0880	-1.209	403		-0.776		Reconstructed
P2	Extended	C1 (Lake Edge)	0.1004	-0.487	350	-0.205	-0.073	132	Calibration
CEH24	Reference	C2 (Dune)	0.0745	-0.441	447	0.015	-0.029	-44	Calibration
CEH28	Reference	C2 (Dune)	0.0774	-0.619	415	-0.128	-0.202	-74	Calibration
D10	Reference	C2 (Dune)	0.0595	-0.487	447	-0.050	-0.074	-24	Calibration
D17	Reference	C2 (Dune)	0.0724	-0.430	485	-0.035	-0.017	18	Calibration
D41	Reference	C2 (Dune)	0.0713	-0.507	326	-0.097	-0.093	4	Calibration
D43	Reference	C2 (Dune)	0.0770	-0.667	312	-0.207	-0.249	-41	Calibration
D44	Reference	C2 (Dune)	0.0936	-0.758	296	-0.305	-0.337	-31	Calibration
D5	Reference	C2 (Dune)	0.0739	-0.619	327	-0.168	-0.201	-33	Calibration
D6	Reference	C2 (Dune)	0.0672	-0.602	397	-0.138	-0.185	-47	Calibration
D7	Reference	C2 (Dune)	0.0790	-0.578	298	-0.108	-0.162	-54	Calibration
D8	Reference	C2 (Dune)	0.0568	-0.456	491	-0.058	-0.043	15	Calibration
L7	Reference	C2 (Dune)	0.0853	-1.624	222	-1.137	-1.180	-43	Calibration
NW3	Reference	C2 (Dune)	0.0752	-0.684	330	-0.178	-0.265	-87	Calibration
NW4	Reference	C2 (Dune)	0.0592	-0.497	532	-0.102	-0.083	19	Calibration
NW4B	Reference	C2 (Dune)	0.0611	-0.489	530	-0.118	-0.075	43	Calibration
T41B	Reference	C2 (Dune)	0.0658	-0.414	390	-0.053	-0.002	51	Calibration
T41C	Reference	C2 (Dune)	0.0708	-0.450	387	-0.073	-0.037	36	Calibration
T41D	Reference	C2 (Dune)	0.0690	-0.680	353	-0.269	-0.261	8	Calibration
WMC1	Reference	C2 (Dune)	0.0582	-0.621	489	-0.235	-0.203	31	Calibration
CEH37	Extended	C2 (Dune)	0.0807	-0.602	291	0.037	-0.185	-222	Calibration
CEH7	Extended	C2 (Dune)	0.0536	-0.819	421	-0.162	-0.396	-234	Calibration
CEH8	Extended	C2 (Dune)	0.0334	-0.746	487	-0.299	-0.325	-26	Calibration
PE	Extended	C2 (Dune)	0.0723	-0.878	405	-0.511	-0.454	58	Calibration
CEH1	Reference	C3 (Western Residual)	0.0666	-0.682	413	-0.165	-0.263	-98	Calibration
CEH10	Reference	C3 (Western Residual)	0.1086	-1.345	280	-0.880	-0.908	-28	Calibration
CEH18	Reference	C3 (Western Residual)	0.0616	-0.966	393	-0.586	-0.540	46	Calibration
CEH21	Reference	C3 (Western Residual)	0.0437	-1.157	389	-0.848	-0.725	123	Calibration
CEH36	Reference	C3 (Western Residual)	0.0506	-0.722	429	-0.393	-0.302	91	Calibration
CEH39	Reference	C3 (Western Residual)	0.0335	-0.521	681	-0.174	-0.106	68	Calibration
CEH4	Reference	C3 (Western Residual)	0.0691	-0.848	290	-0.493	-0.425	68	Calibration
CEH40	Reference	C3 (Western Residual)	0.0371	-0.646	749	-0.309	-0.228	81	Calibration
CEH41	Reference	C3 (Western Residual)	0.0373	-0.535	705	-0.143	-0.120	22	Calibration
CEH42	Reference	C3 (Western Residual)	0.0518	-0.504	515	-0.082	-0.090	-8	Calibration
CEH9	Reference	C3 (Western Residual)	0.0350	-0.725	647	-0.371	-0.305	67	Calibration
D15	Reference	C3 (Western Residual)	0.0628	-0.423	417	0.055	-0.011	-65	Calibration
D25	Reference	C3 (Western Residual)	0.0658	-0.627	412	-0.133	-0.209	-76	Calibration
D38	Reference	C3 (Western Residual)	0.1089	-0.726	283	-0.257	-0.305	-49	Calibration
D9	Reference	C3 (Western Residual)	0.0623	-0.566	445	-0.130	-0.150	-20	Calibration
NW1	Reference	C3 (Western Residual)	0.1002	-1.626	281	-1.165	-1.182	-16	Calibration
NW11	Reference	C3 (Western Residual)	0.0756	-0.862	355	-0.382	-0.438	-56	Calibration
NW13	Reference	C3 (Western Residual)	0.0885	-0.799	301	-0.334	-0.377	-43	Calibration
NW2	Reference	C3 (Western Residual)	0.0816	-0.849	350	-0.348	-0.426	-78	Calibration
NW5	Reference	C3 (Western Residual)	0.0493	-0.741	535	-0.290	-0.320	-30	Calibration

Well	Network	Cluster	β_3 (mth^{-1})	EWI (m bg)	SE (mm)	MSL5 obs	MSL5 recon	Resid. (mm)	Status
NW6	Reference	C3 (Western Residual)	0.0660	-0.629	314	-0.199	-0.212	-13	Calibration
NW7	Reference	C3 (Western Residual)	0.0506	-1.374	374	-0.923	-0.937	-14	Calibration
T41A	Reference	C3 (Western Residual)	0.0963	-0.492	377	0.017	-0.078	-95	Calibration
WMC2	Reference	C3 (Western Residual)	0.0474	-0.657	499	-0.233	-0.239	-6	Calibration
WMC3	Reference	C3 (Western Residual)	0.0457	-1.249	480	-0.877	-0.815	62	Calibration
WMC4	Reference	C3 (Western Residual)	0.0889	-0.831	287	-0.369	-0.407	-39	Calibration
CEH22	Extended	C3 (Western Residual)	0.0560	-0.751	223	-0.473	-0.330	144	Calibration
CEH35	Extended	C3 (Western Residual)	0.0652	-0.799	394	-0.552	-0.377	175	Calibration
CEH38	Extended	C3 (Western Residual)	0.0600	-0.813	292	-0.499	-0.390	109	Calibration
P1	Extended	C3 (Western Residual)	0.0467	-0.290	698		0.118		Reconstructed
PW	Extended	C3 (Western Residual)	0.0550	-0.772	450	-0.402	-0.351	51	Calibration
CEH2	Reference	C4 (Main Forest)	0.0194	-1.206	1408	-0.733	-0.773	-40	Out of scope
CEH20	Reference	C4 (Main Forest)	0.0266	-1.215	882	-0.857	-0.782	75	Out of scope
CEH30	Reference	C4 (Main Forest)	0.0259	-1.665	831	-1.353	-1.219	134	Out of scope
CEH32	Reference	C4 (Main Forest)	0.0163	-1.507	1658	-1.281	-1.066	216	Out of scope
CEH33	Reference	C4 (Main Forest)	0.0226	-1.636	1014	-1.350	-1.192	158	Out of scope
CEH34	Reference	C4 (Main Forest)	0.0094	-0.519	2574	-0.411	-0.104	307	Out of scope
NW10	Reference	C4 (Main Forest)	0.0394	-1.049	618	-0.560	-0.620	-60	Out of scope
CEH15	Extended	C4 (Main Forest)	0.0354	-1.373	538	-0.698	-0.936	-237	Out of scope
FE1	Extended	C4 (Main Forest)	0.0156	-0.906	1298	-0.355	-0.481	-126	Out of scope
LIS1	Extended	C4 (Main Forest)	0.0274	-0.948	1046	-0.319	-0.521	-203	Out of scope
CEH16	Reference	C5 (Coastal Forest)	0.0390	-0.960	509	-0.646	-0.534	112	Out of scope
CEH17	Reference	C5 (Coastal Forest)	0.0605	-2.179	277	-1.893	-1.719	174	Out of scope
CEH19	Reference	C5 (Coastal Forest)	0.0445	-1.029	400	-0.761	-0.601	161	Out of scope
CEH31	Reference	C5 (Coastal Forest)	0.0405	-1.614	616	-1.300	-1.170	130	Out of scope
NW9	Reference	C5 (Coastal Forest)	0.0508	-0.913	415	-0.592	-0.488	104	Out of scope
CEH3	Extended	C5 (Coastal Forest)	0.0559	-1.168	256	-1.027	-0.736	291	Out of scope
FE2	Extended	C5 (Coastal Forest)	0.0473	-1.713	366	-1.269	-1.266	3	Out of scope
FE3	Extended	C5 (Coastal Forest)	0.0627	-2.031	316	-1.641	-1.575	66	Out of scope
FE4	Extended	C5 (Coastal Forest)	0.0475	-1.615	475	-1.153	-1.171	-18	Out of scope
NW8	Extended	C5 (Coastal Forest)	0.0522	-1.462	354	-1.322	-1.022	300	Out of scope
NW8B	Extended	C5 (Coastal Forest)	0.0609	-1.588	325	-1.125	-1.145	-20	Out of scope

S7.2 Datum offsets against the van Willegen quadrats

Van Willegen et al. (2025) publish their spring levels per vegetation quadrat, referenced to each quadrat's ground surface. Their series are this network's readings re-referenced, so the pipeline's five-year mean spring level at the piezometer reproduces each quadrat's series after one constant offset: the quadrat's height relative to the dipwell's ground datum. Table S7.2 lists those offsets. Anyone comparing a level in this report with one referenced to quadrat ground needs them, and Note S12 uses them to carry piezometer metrics to the quadrats.

Table S7.2. Fixed datum offset between each van Willegen et al. (2025) vegetation quadrat's ground surface and its dipwell's, from the pipeline's five-year mean spring level (MSL5) against their published per-quadrat series at the same piezometer, paired on window-end year (n = window-ends). Their series are this network's readings re-referenced to quadrat ground, so the offset is the quadrat's height above (negative: below) the dipwell's ground datum; offset = mean of pipeline minus published. Residual MAD is after the offset. 96 quadrats at 21 piezometers, 614 pairs; offsets -403 to +413 mm, up to 497 mm of relief between one dipwell's quadrats; residual MAD 0.6 mm. A level quoted in this report cannot be compared with one referenced to quadrat ground without applying that quadrat's offset. Source: 26_table_s7_2_vw_datum_offsets.csv.

Quadrat	Piezometer	Cluster	n	Datum offset (mm)	Residual MAD (mm)
CEH1-1	CEH1	C3 (Western Residual)	9	-104	0.0
CEH1-2	CEH1	C3 (Western Residual)	9	240	0.0
CEH1-3	CEH1	C3 (Western Residual)	9	75	0.0
CEH1-4	CEH1	C3 (Western Residual)	9	93	0.0
CEH1-5	CEH1	C3 (Western Residual)	9	-134	0.0
CEH1-6	CEH1	C3 (Western Residual)	9	-191	0.0
CEH22-1	CEH22	Extended	6	-47	0.0
CEH22-2	CEH22	Extended	6	202	0.0
CEH22-3	CEH22	Extended	6	69	0.0
CEH22-4	CEH22	Extended	6	372	0.0
CEH22-5	CEH22	Extended	6	124	0.0
CEH22-6	CEH22	Extended	6	74	0.0
CEH22-7	CEH22	Extended	6	-87	0.0
CEH23-1	CEH23	C1 (Lake Edge)	2	-67	0.0
CEH23	CEH23	C1	2	-15	0.0

-2		(Lake Edge)			
CEH23-3	CEH23	C1 (Lake Edge)	2	52	0.0
CEH23-4	CEH23	C1 (Lake Edge)	2	-22	0.0
CEH23-5	CEH23	C1 (Lake Edge)	2	51	0.0
CEH24-1	CEH24	C2 (Dune)	5	-34	0.0
CEH24-2	CEH24	C2 (Dune)	5	-100	0.0
CEH24-4	CEH24	C2 (Dune)	5	-207	0.0
CEH24-5	CEH24	C2 (Dune)	5	-146	0.0
CEH26-1	CEH26	C1 (Lake Edge)	5	-32	0.0
CEH26-2	CEH26	C1 (Lake Edge)	5	-42	0.0
CEH26-3	CEH26	C1 (Lake Edge)	5	16	0.0
CEH26-4	CEH26	C1 (Lake Edge)	5	66	0.0
CEH26-5	CEH26	C1 (Lake Edge)	5	-71	0.0
CEH4-1	CEH4	C3 (Western Residual)	9	-73	0.0
CEH4-2	CEH4	C3 (Western Residual)	9	59	0.0
CEH4-3	CEH4	C3 (Western Residual)	9	81	0.0
CEH5-1	CEH5	C1 (Lake	9	-43	3.2

		Edge)			
CEH5-2	CEH5	C1 (Lake Edge)	9	-30	3.2
CEH5-3	CEH5	C1 (Lake Edge)	9	-40	3.2
CEH5-4	CEH5	C1 (Lake Edge)	9	-95	3.2
CEH5-5	CEH5	C1 (Lake Edge)	9	-39	3.2
CEH6-1	CEH6	C1 (Lake Edge)	9	64	0.0
CEH6-2	CEH6	C1 (Lake Edge)	9	157	0.0
CEH6-3	CEH6	C1 (Lake Edge)	9	77	0.0
CEH8-1	CEH8	Extended	9	154	2.6
CEH8-2	CEH8	Extended	9	-27	2.6
CEH8-3	CEH8	Extended	9	375	2.6
CEH9-1	CEH9	C3 (Western Residual)	4	-80	0.0
CEH9-10	CEH9	C3 (Western Residual)	4	-88	0.0
CEH9-2	CEH9	C3 (Western Residual)	4	-217	0.0
CEH9-3	CEH9	C3 (Western Residual)	4	-384	0.0
CEH9-4	CEH9	C3 (Western	4	-403	0.0

		Residual)			
CEH9-5	CEH9	C3 (Western Residual)	4	-402	0.0
CEH9-6	CEH9	C3 (Western Residual)	4	5	0.0
CEH9-7	CEH9	C3 (Western Residual)	4	94	0.0
CEH9-8	CEH9	C3 (Western Residual)	4	-130	0.0
CEH9-9	CEH9	C3 (Western Residual)	4	-106	0.0
NW2-1	NW2	C3 (Western Residual)	9	169	0.0
NW2-2	NW2	C3 (Western Residual)	9	383	0.0
NW2-3	NW2	C3 (Western Residual)	9	288	0.0
NW2-4	NW2	C3 (Western Residual)	9	44	0.0
NW2-5	NW2	C3 (Western Residual)	9	-78	0.0
NW3-1	NW3	C2 (Dune)	4	70	0.0

NW3-2	NW3	C2 (Dune)	4	-19	0.0
NW3-3	NW3	C2 (Dune)	4	33	0.0
NW3-4	NW3	C2 (Dune)	4	34	0.0
NW4-1	NW4	C2 (Dune)	9	20	0.0
NW4-10	NW4	C2 (Dune)	9	203	0.0
NW4-2	NW4	C2 (Dune)	9	-23	0.0
NW4-3	NW4	C2 (Dune)	9	-88	0.0
NW4-4	NW4	C2 (Dune)	9	-32	0.0
NW4-5	NW4	C2 (Dune)	9	-50	0.0
NW4-6	NW4	C2 (Dune)	9	43	0.0
NW4-7	NW4	C2 (Dune)	9	84	0.0
NW4-8	NW4	C2 (Dune)	9	147	0.0
NW4-9	NW4	C2 (Dune)	9	234	0.0
NW5-1	NW5	C3 (Western Residual)	9	7	0.0
NW5-2	NW5	C3 (Western Residual)	9	-31	0.0
NW5-3	NW5	C3 (Western Residual)	9	-54	0.0
NW5-4	NW5	C3 (Western Residual)	9	29	0.0
NW5-5	NW5	C3 (Western Residu	9	65	0.0

		al)			
NW6-1	NW6	C3 (Western Residual)	4	87	0.0
NW6-2	NW6	C3 (Western Residual)	4	207	0.0
NW6-3	NW6	C3 (Western Residual)	4	5	0.0
NW6-4	NW6	C3 (Western Residual)	4	68	0.0
NW6-5	NW6	C3 (Western Residual)	4	-42	0.0
NW7-1	NW7	C3 (Western Residual)	4	143	0.0
NW7-2	NW7	C3 (Western Residual)	4	-11	0.0
NW7-3	NW7	C3 (Western Residual)	4	-19	0.0
NW7-4	NW7	C3 (Western Residual)	4	-19	0.0
NW7-5	NW7	C3 (Western Residual)	4	-15	0.0
T41A-4	T41A	C3 (Western	5	88	0.0

		Residual)			
T41B-2	T41B	C2 (Dune)	5	74	4.8
T41B-5	T41B	C2 (Dune)	5	5	4.8
T41B-7	T41B	C2 (Dune)	5	191	4.8
T41B-8	T41B	C2 (Dune)	5	413	4.8
T41C-3	T41C	C2 (Dune)	5	48	0.0
T41D-1	T41D	C2 (Dune)	5	24	0.0
T41D-6	T41D	C2 (Dune)	5	-173	0.0
WMC2-1	WMC2	C3 (Western Residual)	7	30	3.8
WMC2-2	WMC2	C3 (Western Residual)	7	10	3.8
WMC2-3	WMC2	C3 (Western Residual)	7	4	3.8

S7.3 Community levels at the quadrats

Section 4.8.3 of the report reads the dune-slack community levels where the slack floor is surveyed: at van Willegen et al.'s quadrats, each on its own ground. Table S7.3 lists, for each quadrat, the four-year mean annual minimum on quadrat ground over Curreli et al.'s window and over the latest window, with its band against the SD15b and SD16 community means.

Table S7.3. Four-year mean annual minimum water level at each van Willegen et al. (2025) vegetation quadrat, on the quadrat's own ground (the piezometer's level less the quadrat's Table S7.2 offset; m, negative below ground), over Curreli et al.'s window (2007–10) and the latest (2022–25), with its band against the Curreli et al. (2013) community means, SD15b -0.61 m and SD16 -0.98 m. 47 quadrats are read in both windows; every one is deeper in 2022–25. — where a quadrat's piezometer has no complete window. (Source: 26_curreli_min_quadrat_bands.csv).

Quadrat	Piezometer	Cluster	2007–10 (m)	2022–25 (m)	Band (2007–10 → 2022–25)
CEH1-1	CEH1	C3	−0.73	−0.90	SD15b–SD16 → SD15b–SD16
CEH1-2	CEH1	C3	−1.08	−1.24	below SD16 → below SD16

CEH1-3	CEH1	C3	−0.91	−1.08	SD15b–SD16 → below SD16
CEH1-4	CEH1	C3	−0.93	−1.10	SD15b–SD16 → below SD16
CEH1-5	CEH1	C3	−0.70	−0.87	SD15b–SD16 → SD15b–SD16
CEH1-6	CEH1	C3	−0.65	−0.81	SD15b–SD16 → SD15b–SD16
CEH23-1	CEH23	C1	—	−0.81	— → SD15b–SD16
CEH23-2	CEH23	C1	—	−0.87	— → SD15b–SD16
CEH23-3	CEH23	C1	—	−0.93	— → SD15b–SD16
CEH23-4	CEH23	C1	—	−0.86	— → SD15b–SD16
CEH23-5	CEH23	C1	—	−0.93	— → SD15b–SD16
CEH24-1	CEH24	C2	—	−0.80	— → SD15b–SD16
CEH24-2	CEH24	C2	—	−0.73	— → SD15b–SD16
CEH24-4	CEH24	C2	—	−0.62	— → SD15b–SD16
CEH24-5	CEH24	C2	—	−0.68	— → SD15b–SD16
CEH26-1	CEH26	C1	—	−0.84	— → SD15b–SD16
CEH26-2	CEH26	C1	—	−0.83	— → SD15b–SD16
CEH26-3	CEH26	C1	—	−0.89	— → SD15b–SD16
CEH26-4	CEH26	C1	—	−0.94	— → SD15b–SD16
CEH26-5	CEH26	C1	—	−0.80	— → SD15b–SD16
CEH4-1	CEH4	C3	−0.52	−0.99	above SD15b → below SD16
CEH4-2	CEH4	C3	−0.65	−1.12	SD15b–SD16 → below SD16
CEH4-3	CEH4	C3	−0.67	−1.15	SD15b–SD16 → below SD16
CEH5-1	CEH5	C1	−0.70	−0.92	SD15b–SD16 → SD15b–SD16
CEH5-2	CEH5	C1	−0.72	−0.94	SD15b–SD16 → SD15b–SD16
CEH5-3	CEH5	C1	−0.71	−0.93	SD15b–SD16 → SD15b–SD16
CEH5-4	CEH5	C1	−0.65	−0.87	SD15b–SD16 → SD15b–SD16
CEH5-5	CEH5	C1	−0.71	−0.93	SD15b–SD16 → SD15b–SD16
CEH6-1	CEH6	C1	−0.71	−0.81	SD15b–SD16 → SD15b–SD16
CEH6-2	CEH6	C1	−0.81	−0.90	SD15b–SD16 → SD15b–SD16
CEH6-3	CEH6	C1	−0.73	−0.82	SD15b–SD16 → SD15b–SD16
CEH8-1	CEH8	C2	−0.91	—	SD15b–SD16 → —

CEH8-2	CEH8	C2	−0.73	—	SD15b–SD16 → —
CEH8-3	CEH8	C2	−1.13	—	below SD16 → —
CEH9-1	CEH9	C3	−0.76	−1.00	SD15b–SD16 → below SD16
CEH9-2	CEH9	C3	−0.63	−0.86	SD15b–SD16 → SD15b–SD16
CEH9-3	CEH9	C3	−0.46	−0.70	above SD15b → SD15b–SD16
CEH9-4	CEH9	C3	−0.44	−0.68	above SD15b → SD15b–SD16
CEH9-5	CEH9	C3	−0.44	−0.68	above SD15b → SD15b–SD16
CEH9-6	CEH9	C3	−0.85	−1.09	SD15b–SD16 → below SD16
CEH9-7	CEH9	C3	−0.94	−1.17	SD15b–SD16 → below SD16
CEH9-8	CEH9	C3	−0.71	−0.95	SD15b–SD16 → SD15b–SD16
CEH9-9	CEH9	C3	−0.74	−0.97	SD15b–SD16 → SD15b–SD16
CEH9-10	CEH9	C3	−0.75	−0.99	SD15b–SD16 → below SD16
NW2-1	NW2	C3	−1.13	−1.31	below SD16 → below SD16
NW2-2	NW2	C3	−1.35	−1.52	below SD16 → below SD16
NW2-3	NW2	C3	−1.25	−1.43	below SD16 → below SD16
NW2-4	NW2	C3	−1.01	−1.18	below SD16 → below SD16
NW2-5	NW2	C3	−0.88	−1.06	SD15b–SD16 → below SD16
NW3-1	NW3	C2	−0.86	—	SD15b–SD16 → —
NW3-2	NW3	C2	−0.77	—	SD15b–SD16 → —
NW3-3	NW3	C2	−0.82	—	SD15b–SD16 → —
NW3-4	NW3	C2	−0.82	—	SD15b–SD16 → —
NW4-1	NW4	C2	−0.58	−0.77	above SD15b → SD15b–SD16
NW4-2	NW4	C2	−0.54	−0.73	above SD15b → SD15b–SD16
NW4-3	NW4	C2	−0.47	−0.66	above SD15b → SD15b–SD16
NW4-4	NW4	C2	−0.53	−0.72	above SD15b → SD15b–SD16
NW4-5	NW4	C2	−0.51	−0.70	above SD15b → SD15b–SD16
NW4-6	NW4	C2	−0.60	−0.80	above SD15b → SD15b–SD16
NW4-7	NW4	C2	−0.64	−0.84	SD15b–SD16 → SD15b–SD16
NW4-8	NW4	C2	−0.71	−0.90	SD15b–SD16 → SD15b–SD16
NW4-9	NW4	C2	−0.79	−0.99	SD15b–SD16 → below SD16
NW4-10	NW4	C2	−0.76	−0.96	SD15b–SD16 → SD15b–SD16
NW5-1	NW5	C3	−0.83	−1.03	SD15b–SD16 → below SD16
NW5-2	NW5	C3	−0.79	−0.99	SD15b–SD16 → below SD16
NW5-3	NW5	C3	−0.77	−0.97	SD15b–SD16 → SD15b–SD16
NW5-4	NW5	C3	−0.85	−1.05	SD15b–SD16 → below SD16
NW5-5	NW5	C3	−0.89	−1.09	SD15b–SD16 → below SD16
NW6-1	NW6	C3	−0.72	—	SD15b–SD16 → —

NW6-2	NW6	C3	−0.84	—	SD15b–SD16 → —
NW6-3	NW6	C3	−0.64	—	SD15b–SD16 → —
NW6-4	NW6	C3	−0.70	—	SD15b–SD16 → —
NW6-5	NW6	C3	−0.59	—	above SD15b → —
NW7-1	NW7	C3	−1.52	—	below SD16 → —
NW7-2	NW7	C3	−1.37	—	below SD16 → —
NW7-3	NW7	C3	−1.36	—	below SD16 → —
NW7-4	NW7	C3	−1.36	—	below SD16 → —
NW7-5	NW7	C3	−1.36	—	below SD16 → —
T41A-4	T41A	C3	—	−1.01	— → below SD16
T41B-2	T41B	C2	—	−0.86	— → SD15b–SD16
T41B-5	T41B	C2	—	−0.79	— → SD15b–SD16
T41B-7	T41B	C2	—	−0.97	— → SD15b–SD16
T41B-8	T41B	C2	—	−1.20	— → below SD16
T41C-3	T41C	C2	—	−0.88	— → SD15b–SD16
T41D-1	T41D	C2	—	−1.03	— → below SD16
T41D-6	T41D	C2	—	−0.83	— → SD15b–SD16
WMC2-1	WMC2	C3	—	−1.00	— → below SD16
WMC2-2	WMC2	C3	—	−0.98	— → SD15b–SD16
WMC2-3	WMC2	C3	—	−0.97	— → SD15b–SD16

S7.4 Reproducibility

The equilibrium wetness index, its calibration onto the MSL5 scale, and this table are computed by 26_van_willegen_msl.py. Reference-tier coefficients are read from 03_master_data.csv (Script 03) and extended-tier coefficients are fitted in the same script using the shared state-space routine in utils/model_utils.py. The per-well index and its standard errors are exported to 26_equilibrium_wetness_index_per_well.csv, the observed-versus-reconstructed comparison to 26_ewi_msl5_comparison.csv, and the calibration constants to 26_report_numbers.csv. Table S7.1 is emitted directly from those outputs as 26_table_s7_1_ewi_per_well.csv rather than transcribed, so it cannot drift from them; Table S7.2 is emitted by the same script (Pass 7b) as 26_table_s7_2_vw_datum_offsets.csv, and Table S7.3 (Pass 7c) as 26_table_s7_3_quadrat_community_levels.csv.

Supplementary Note S8: Spring-Mean (MAM) Seasonal Robustness Analysis

S8.1 Purpose

The site's ecologically framed BACI and coastal-gradient analyses use the annual summer minimum (Jun–Sep) as the response metric — the drought-stress quantity that determines dune-slack habitat viability. This note tests whether that choice drives the conclusions, by re-running the seasonal analyses on the annual spring mean (Mar–May, the season the Curreli ecological thresholds are defined on) as a second metric through identical code paths. Spring is added, never substituted: the summer minimum remains the headline. The spring mean carries the same N as the summer minimum (one value per well-year) and, being a three-month mean rather than an extreme-order statistic, is the less noisy of the two. The spring window and its strict 3-of-3 completeness rule are the config constants MSL_SPRING_MONTHS = (3, 4, 5) and MSL_MIN_MONTHS_PER_SPRING = 3, shared with the van Willegen MSL analyses (Methods Supplement F.4).

S8.2 Scraping Intervention (Script 09c)

At the scraped well CEH36, the spring mean shifts post-scraping by +95 mm against the paired CEH4 control (p = 0.022) and +19 mm against the climate centroid (p = 0.79, not significant), versus +195 mm and +161 mm respectively for the summer minimum. Scraping raises the spring level too, by less than it raises the summer minimum, and the two controls — which agreed in summer — disagree in spring because every non-impact well also shifts negative against the climate centroid, the signature of the regional spring centroid itself having risen rather than of the impact wells falling. With n_pre = 4 years these p-values are fragile.

S8.3 Clearfell BACI (Scripts 10d, 10l)

The clearfell tier-mean spring shifts against the Forest control (Script 10d) are Impact −2, Edge −107, Forest-control −9 and Climate-control −63 mm, none individually significant. The four-zone spring model (Script 10l; 14-year panel, $R^2 = 0.59$) returns the differential felling steps in Table S8.1. The Impact (−2 mm, p = 0.88) and C3/Warren (−11 mm, p = 0.56) steps are not significant, and the Impact step agrees with the per-well comparison against the same control (WMC3 −2 mm, p = 0.97; Script 10d). The Edge step is −53 mm (p = 0.027): negative, and so not a felling benefit; it is consistent with the maturing-replant confound at the Edge wells (main report Section 4.6.8) and rests on cluster-robust standard errors over very few clusters, so it is reported for completeness, not as a felling effect. In summer the felling signal sits at the Impact zone in the monthly ANCOVA, and neither seasonal branch of the four-zone model resolves a positive felling step. The spring panel spans 14 years (2011 + 2013–2025) against the summer panel's 13, because WMC3's Mar–May record is near-complete and 2019 — unusable in summer — is fully measured in spring, even under the stricter 3-of-3 rule.

Table S8.1. Four-zone differential felling step relative to the Forest control zone, spring mean versus summer minimum (Scripts 10l_06, 10l_01). Steps in mm; C3/Warren is the second control zone, expected near zero.

Zone	Spring step ± SE (mm)	Spring p	Summer step ± SE (mm)	Summer p
Impact	−2.3 ± 15.7	0.885	−7.3 ± 7.9	0.358
Edge	−53.2 ±	0.027	−28.6 ± 20.5	0.162

	24.0			
C3/Warren	−10.6 ± 18.4	0.563	+12.0 ± 10.6	0.257

S8.4 Cluster Trends and the Coastal Gradient (Scripts 14, 25)

The spring-mean cluster-centroid trend (Script 14) is flat for every cluster except C5 Coastal Forest (−0.038 m yr⁻¹, p = 0.021); C1–C4 lie between −0.004 and +0.008 m yr⁻¹, none significant. The contrast with the summer minimum is the substantive result (Table S8.2, Figure S8.1): summer minima decline across every cluster (−7 to −36 mm yr⁻¹), but spring means are essentially stable inland — the inland summer-minimum decline is a growing-season phenomenon, not a fall in the baseline water table. C5 is the exception: it declines in both seasons (fixed-effects cluster summer −29.1, spring −23.5 mm yr⁻¹; on the balanced basis the decomposition uses, −31.98 and −23.91 mm yr⁻¹), and applying the all-season coastal-retreat gradient (Script 25, Methods Supplement S.15) to the spring metric attributes most of C5's spring decline to coastal retreat (coastal gradient −17.07 mm yr⁻¹ against a balanced basis of −23.91 mm yr⁻¹, or 71 %, which leaves −9.50 mm yr⁻¹ unexplained); at a four-well cluster the share is not resolvable more finely. The coastal term is present in both seasons, consistent with the view that C5's decline is a year-round coastal-boundary effect rather than a drought-season artefact. What the decomposition does not contain is a far-field climate rate. The panel carries a constant term c_far alongside a cumulative-water-balance (CWB) covariate, and although the CWB carries a trend over the panel's own fitted span, the two are separately identified: the variance inflation factor of c_far against the covariate is 1.01 (Methods Supplement S.15). What disqualifies the constant is not confounding but instability. Refitting the same specification over a different window moves the constant across a range of some 24 mm yr⁻¹ with the well set held fixed (25_12_window_sweep.csv) and reverses its sign — the committed matched-window sensitivity (25_11_matched_window_sensitivity.csv, emitted for inspection and not adopted) returns c_far between −8.39 and +6.65 mm yr⁻¹ and their sum between −5.31 and +9.48 mm yr⁻¹ — and the MAM-only refit in 25_01_panel_fit_parameters.csv shifts it again with the well set unchanged. The constant is therefore a property of the window the panel is fitted over, not a rate: no far-field climate background should be read from it in either direction. What the fit does support is the distance dependence. The coast-edge anomaly δ_0 and the inland reach L_cg are identified by the shape of the gradient rather than by its level, and it is those two, not the constant, on which the C5 attribution above rests; the matched-window sensitivity moves them as well, which is why it is reported rather than adopted.

The full-panel season × δ(d)·t interaction test (Methods Supplement S.15) qualifies this at the gradient level: on the forest-free panel (10,909 observations, one model) the spring modulation of the gradient drift rate is γ = +0.309 ± 0.060 (t = 5.16, p < 0.001) for the linear-capped form and +0.304 ± 0.059 (t = 5.15, p < 0.001) for the exponential. Both forms reject season-independence, and both put γ above zero: the coastal-retreat drift rate is about 31 % steeper in Mar–May than over the rest of the year. That the estimate is significant, of the same sign and of compatible magnitude under both decay forms indicates a seasonal modulation of the gradient itself rather than an artefact of the assumed functional form. The MAM-only refit of the gradient panel (a sensitivity block in 25_01_panel_fit_parameters.csv) points the same way: it loses power as expected — the coast-edge anomaly δ_0 fitted standard error roughly doubles (1.97 to 3.72 mm yr⁻¹) — but returns a steeper spring coast-edge anomaly than the all-season fit (δ_0 −34.06 against −31.28 mm yr⁻¹, L_cg 894 and 902 m), a change of about 9 % in the same direction as γ. The all-season gradient is therefore a season-averaged quantity, not a season-independent one. Where it is applied unchanged to a spring metric — as in the C5 decomposition above and in Table S8.2 — it understates the spring coastal contribution, and those spring coastal-gradient figures should be read as lower bounds.

Table S8.2. Observed cluster slope on the declared balanced basis (observed_balanced_annual_mean_mm_yr, named in the file's own decomposition_basis column) — the slope of the annual cross-well mean of the per-well seasonal metric — summer minimum versus spring mean, with the spring-mean decomposition under the all-season forest-free linear-capped gradient (Scripts 25_08, 25_03_cluster_partition_spring). Slopes in mm yr^{-1}. The cluster-centroid slopes from Script 14 are retained as context columns (observed_centroid_mm_yr, beside observed_per_well_mean_mm_yr) in 25_03_cluster_partition_spring.csv and are quoted in the text above. Beyond the coastal gradient the modelled total carries two cluster-independent terms, both taken from the committed file: the contribution of the cumulative-water-balance covariate (climate_cwb_mm_yr) and the fitted far-field asymptote c_far (far_field_offset_mm_yr). The asymptote is a window statistic rather than a rate — the fit recovers only their sum — and that sum is a property of the fit window rather than a rate (S8.4), so neither is reported as a climate background. The unexplained column (unexplained_mm_yr) is the basis less the coastal gradient less that sum. The gradient applied here is the all-season fit, and the interaction test reported above finds the gradient significantly steeper in spring (γ = +0.309, $p < 0.001$), so the spring coastal-gradient column is a lower bound on the spring coastal contribution rather than a point estimate of it.

Cluster	Summer obs (basis)	Spring obs (basis)	Spring coastal gradient	Spring unexplained
C1 (Lake Edge)	−11.0	+7.9	−0.6	+5.8
C2 (Dune)	−14.6	+0.6	0.0	−2.1
C3 (Western Residual)	−7.5	+3.6	−7.0	+8.0
C4 (Main Forest)	−6.5	+11.9	−4.8	+14.1
C5 (Coastal Forest)	−32.0	−23.9	−17.1	−9.5

Figure S8.1 (25_08_spring_vs_summer_comparison.png) shows the observed centroid slope by cluster, summer minimum versus spring mean, with the all-season gradient marked. Figure S8.2 (14_climate_trajectory_spring.png) shows the observed spring-mean trajectories by cluster with per-cluster OLS trends.

S8.5 Conclusion

The spring-mean analysis confirms most of the summer-minimum conclusions and revises one. Scraping raises spring levels as it does summer minima; the qualitative clearfell-BACI picture does not depend on the choice of summer minimum over spring mean; and the inland summer-minimum decline is growing-season specific (spring flat). The revision concerns the coastal-retreat gradient. The C5 signal is present year-round, but the season × gradient interaction test finds the gradient itself significantly steeper in spring (γ = +0.135, $p = 0.015$ for the linear-capped form; +0.143, $p = 0.011$ for the exponential), so the all-season gradient is a season-averaged quantity rather than a season-independent one, and applying it unchanged to a spring metric understates the spring coastal contribution. The one nominally significant spring-only result, the Edge four-zone step, is not robust to a single pre-felling year and is not interpreted as a felling effect.

S8.6 Reproducibility

Scripts 09c (v1.5.0), 10d (v1.7.0), 10l (v1.2.0), 14 (v1.4.x), 25 (v1.6.x). Every spring output is additive and the committed summer outputs reproduce byte-identically. The spring season and completeness rule are config.MSL_SPRING_MONTHS and

config.MSL_MIN_MONTHS_PER_SPRING; the interaction test and MAM sensitivity are described in Methods Supplement S.15.

Supplementary Note S9: Drainage-Datum Regime and the Loss Partition

S9.1 Purpose

The SSM drainage term is $\beta_3 \cdot (z_0 + h)$, where z_0 is the drainage datum (report Section 3.4). Because the fitted β_2 and β_3 trade against one another as z_0 moves, the drainage/ET partition of the cluster water balance (report Section 4.2.3, Figure 11) is conditional on the datum. This note documents the datum-regime diagnostic introduced in Script 03 v1.5.0, which shows that the conditionality is confined to shallow datums: the partition is datum-conditional only when the datum lies within the water-table fluctuation range, and stabilises once it does not.

S9.2 Construction

From the committed cluster-centroid datum sweep (03_08_datum_sensitivity.csv; 0.5–8.0 m in 0.1 m steps), Script 03 computes, at each swept datum z_0 and cluster, the fitted drainage flux $\beta_3(D) \cdot (z_0 + \hbar)$ and its share of modelled losses against the atmospheric draw $\beta_2(D) \cdot \overline{PET}$. Here $\hbar$ is the mean end-of-previous-month water level over the aligned SSM frame — the drainage term's own regressor — and $\overline{PET}$ the mean monthly PET over the same rows; both are datum-independent, so the datum dependence of the fluxes comes entirely from the fitted coefficients. The shallow-regime boundary is computed live from the sweep's own β_3 positivity and significance flags (1.5 m under the current record), not hardcoded. Every plotted value is traced in 03_12_partition_vs_datum.csv.

S9.3 Reading

In the shaded shallow regime the datum sits within the water-table fluctuation range: the drainage term reduces to mean-reversion of anomalies, β_3 estimates are unstable or negative, and almost no steady flux is attributed to drainage. Beyond the boundary — the Darcy regime — the term represents outflow above a fixed base: the fitted drainage flux plateaus and the drainage/ET partition stabilises, so the exact choice of datum is second-order. The canonical 3.7 m datum sits on this plateau. Two features corroborate the physical reading. C5 Coastal Forest converges with the open-dune C3 in the Darcy regime despite diverging strongly under shallow datums, consistent with the elevation-dominated coefficient structure of report Section 4.9.4: C5's drainage behaviour is set by its low-lying coastal position rather than its canopy. The unconstrained C4 remains the lone low outlier.
Figure S9.1 (03_12_datum_regime.png). Datum-regime diagnostic: fitted drainage flux $\beta_3 \cdot (z_0 + \hbar)$ (top) and drainage share of modelled losses (bottom) against the reference datum, one line per cluster, with the shallow mean-reversion regime shaded and the canonical 3.7 m datum marked.

S9.4 Reproducibility

Script 03 (v1.5.0); outputs 03_12_partition_vs_datum.csv and 03_12_datum_regime.png. The diagnostic is additive: all pre-existing Script 03 outputs reproduce byte-identically under the v1.5.0 edit. Shares evaluated at the 3.7 m datum reproduce the Table 6 partition of the main report to within about 0.1% (the $\hbar$ window convention).

Supplementary Note S10: Reading the Network-Change Maps (Figures 73–76)

S10.1 Purpose

The four network-change products of report Section 3.8.1 (Figures 73–76) are constructed from the same per-well spring (March–May) series and differ only in climate handling and frame of reference. Because the four constructions are described in necessarily dense parallel prose, this note provides the comparison at a glance: Table S10.1 sets out what each map asks, over what window, and in which frame; Figure S10.1 shows the shared construction as a flow schematic — one source series, two frames (common mode removed versus absolute), four products.

Table S10.1. Construction of the four network-change products. Full construction detail is given in report Section 3.8.1; per-script methodology in the Methods Supplement (chapters S.20).

Map	Question it answers	Window	Climate handling	Frame of reference and units
Figure 73 — Differential movement (Script 32)	Which wells are gaining or losing ground against the site as a whole?	2011–2025	Network-mean spring subtracted each year (common mode removed)	Relative to the site mean; mm yr^{-1}
Figure 74 — Absolute climate-removed trend (Script 36)	How fast is each well itself drifting, once the shared climate signal is removed?	2005–2025	Shared climate signal removed; not re-referenced to the network	Absolute, per well; mm yr^{-1}
Figure 75 — Amplification field (Script 33)	How strongly does each well swing between dry and wet springs?	Dry and wet extreme springs of the record	Common mode removed; expressed as a ratio to the network-mean swing	Relative amplification; dimensionless (×)
Figure 76 — Dry-year spring depth (Script 33)	How close does a dry spring already sit to the ecological thresholds?	Dry extreme springs of the record	None — the absolute dry state is the point	Absolute depth below ground; m

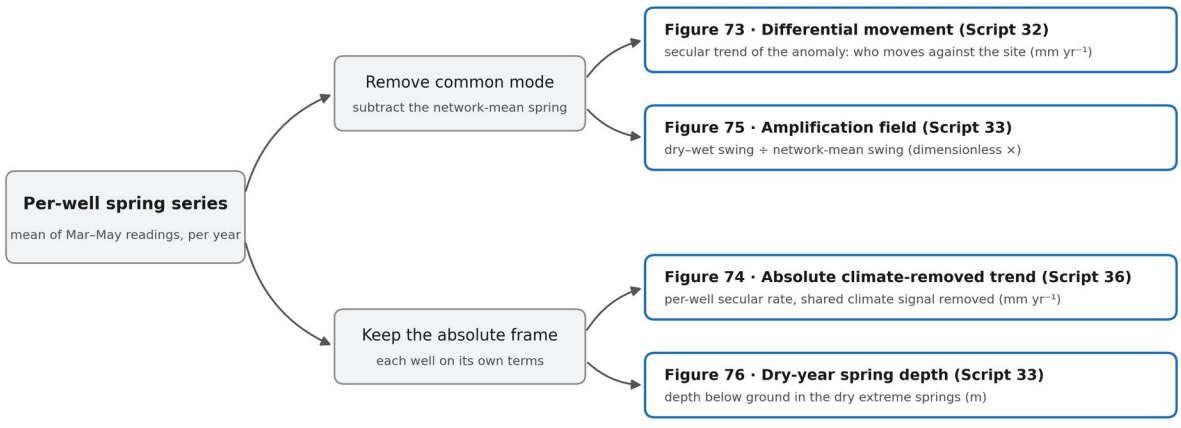


Figure S10.1 (S10_network_change_schematic.png). Construction schematic for the network-change products: the per-well spring series feeds two frames — common mode removed (Figures 73 and 75) and absolute (Figures 74 and 76) — yielding the four maps. A per-well climate-sensitivity coefficient (Script 35) and a two-window sensitivity demonstration (Script 34; Note S11) accompany the group.

S10.2 Sources

Scripts 32 (differential movement), 33 (climate-response envelope and dry-year depth), and 36 (absolute climate-removed trend), with Scripts 34 (Note S11) and 35 as companions. All values trace to the committed per-well CSVs of those scripts.

Supplementary Note S11: Two-Window MSL5 Sensitivity (Script 34)

S11.1 Purpose

The observed MSL5 change map (report Figure 72) differences two five-year spring windows — window-end 2017 against window-end 2023. This note shows directly why that two-window construction cannot resolve absolute site-wide change: the single 2017→2023 comparison sits inside a wide envelope of equally admissible window pairs, so the choice of windows, not the aquifer, sets much of the answer.

S11.2 Method

Per-well MSL5 for a window ending in year Y is the mean of the well's annual spring water level over years Y−4 to Y (from the committed van Willegen aggregation, 26_msl_annual_per_well.csv; wells CEH13 and CEH14 excluded, matching Script 26); a well qualifies for a window only if all five spring-years are present. For each ordered pair of windows the site-mean change is the mean, over the common panel of wells qualifying in both windows, of the per-well change — the panel held fixed across the pair, so composition change cannot inflate the result. A pair is admissible when its common panel reaches at least forty wells, which excludes only the thin seven-well 2005–2009 baseline, so the spread arises from window choice alone among well-sampled windows. All admissible pairs are retained, including those whose current window contains the anomalously wet 2024 spring — admitting them is the point of the demonstration.

S11.3 Result

The 2017→2023 pair gives −105 mm, but this is one interior cell of a wide envelope: across the 55 admissible window pairs the site-mean change spans −0.16 to +0.23 m (−161 to +230 mm) and changes sign — 17 falling, 38 rising — with the most negative pairing 2017→2019 (−161 mm) and the most positive 2015→2024 (+230 mm, resting on the exceptionally wet 2024 spring). A single admissible change of window pair can therefore move the site-mean estimate by more than a third of a metre and reverse its sign, and a trend fitted to the site-mean spring level across the full record is not significantly different from zero (−4.8 mm yr^{-1}, AR-corrected, p = 0.63). On the nine-year mean spring level (MSL9), the handbook's metric, the envelope narrows to −67 to +130 mm over the 36 admissible pairs, 12 falling and 24 rising, but still changes sign, and it does so at every cluster; the first admissible MSL9 window ends in 2017, so every admissible pair shares springs and none compares two separate periods. The window-independent secular signal is recovered instead by the differential-movement and absolute-trend maps (report Figures 73 and 74).
Figure S11.1 (34_window_sensitivity.png). The 2017→2023 MSL5 comparison placed inside the envelope of every admissible window pair; the single reported comparison is one point in a distribution that spans both signs. Source: 34_window_sensitivity.py.

S11.4 Sources

Script 34 (two-window sensitivity), reading the committed per-well annual spring MSL from Script 26 (26_msl_annual_per_well.csv). Window-mean rainfall annotations trace to 00_01_annual_climate_summary.csv. All values trace to committed CSVs.

Supplementary Note S12: Community Reference Values for MSL5 — Specification for Further Work

S12.1 Why the values do not yet exist

Van Willegen et al. (2025) established the five-year mean spring water level as the hydrological metric most strongly related to dune-slack community moisture at Newborough — mean Ellenberg F, with Pearson r of 0.84 to 0.89 across their relevés — and named reference values, or a community "status", on the five-year MSL as the next step. Curreli et al. (2013) supplied community reference values, but for a different quantity: their Table 4 gives, for each National Vegetation Classification community, the mean, standard error and range of the four-year (2006–09) average annual minimum and maximum water level across the quadrats assigned to that community, over a 1 June–31 May hydrological year. The report applies those values to the annual-minimum series (report Sections 4.8.1 and 4.8.3, Table 20, Figure 45b) and not to MSL5. No published source gives a community reference value for a spring mean, and the report does not construct one by transfer. This note specifies how such values would be derived on the same footing as Curreli's, what is already in hand, and the one input that is not.

S12.2 What would be computed

The counterpart of Curreli's Table 4 for the spring metric: for each NVC community, the mean, standard error, range and count of (a) the annual mean spring water level and (b) the five-year mean spring water level at the permanent vegetation quadrats assigned to that community, over a stated period and a stated survey year. Two companion rows would be computed first: the four-year mean annual minimum and the four-year mean annual maximum at the same quadrats over Curreli's period, which must reproduce her Table 4 within its standard errors. That reproduction is the test that the quadrats, the datum treatment and the period are the ones her values rest on; only if it passes do the spring rows inherit the same standing. The product is a set of characteristic values with ranges — where each community was found to sit on the metric — and is reported as such, never as a viability limit.

S12.3 Data in hand

Three of the four inputs exist in the pipeline. First, the monthly water-level record at every van Willegen piezometer, 2005 to date (Script 01), from which Script 26 already computes the annual spring mean, the five-year MSL and, since D-190, the annual minimum and its four-year mean per well. Second, the fixed datum offset between each permanent quadrat's ground surface and its piezometer's, tabulated in Supplementary Table S7.2 (Script 26, Pass 7b): van Willegen's per-quadrat series are this network's readings shifted by that constant, so any hydrological metric at a quadrat is the piezometer's metric plus the offset, and every quadrat-level value in S12.2 is available from the committed outputs without new measurement. Third, the mean Ellenberg F of each relevé in each survey year, from the deposited dataset (van Willegen et al., Mendeley Data), which Script 26 reads for the vegetation cross-validation of report Section 4.8.4.

S12.4 The missing input

The deposited dataset carries no NVC community per quadrat. It is the only input the specification lacks, and it exists in three places. (i) Curreli et al.'s 2010 classification of the 453 relevés, from which Table 4's counts per community are drawn; one column keyed by permanent-quadrat identifier would suffice, and the survey team (Bangor University, UK Centre for Ecology & Hydrology) holds it. (ii) The classification of the repeat relevés of 2012–

2019 by the same surveyor, which would give the community per quadrat per survey year and so identify the quadrats whose community did not change over the decade — the subset on which a multi-year reference value is cleanly defined, since van Willegen et al. showed that communities moved with the hydrology. (iii) As a coarse fallback only, the NVC map of the site (Dargie 1995, National Sand Dune Vegetation Survey; any later resurvey held by Natural Resources Wales) sampled at the quadrat coordinates: a polygon-scale community fifteen years before the relevés, usable to check a classification but not to supply one, because slack-margin quadrats straddle polygon boundaries and communities moved in the interval. Assigning a community from mean Ellenberg F is not an option: Ellenberg F is the quantity MSL5 is calibrated against, and a class derived from it would make the reference values circular.

S12.5 Method

Periods. Two are needed. For the reproduction rows, Curreli's 2006–09 window, which is the pipeline's hydrological years 2007–2010 at these piezometers — the earliest window for which every summer month is present, the record beginning in 2005. For the spring rows, every window-end the record admits, with the community per quadrat taken from the survey year nearest the window-end, or restricted to the stable-community subset of S12.4(ii).
Quadrat values. Each quadrat's metric is its piezometer's metric plus the Table S7.2 offset. Because the offset is a constant it enters every metric additively and leaves the piezometer's admission rules (three of three spring readings for an annual spring mean; every summer month present and the year closed for an annual minimum; every year present for a multi-year mean) unchanged.
Community statistics. For each community, metric and period: the mean, standard error, minimum, maximum and count over quadrats, exactly as Curreli's Table 4 reports them, so that the reproduction rows are comparable cell for cell. Beside them, the count of distinct piezometers in the community and a second standard error computed on piezometer-level means. Quadrats that share a piezometer differ only by a constant and are not independent samples of the hydrology; the quadrat-level standard error is the comparable one and the piezometer-level standard error is the honest one, and the table carries both.
Reproduction test. The four-year mean annual minimum and maximum per community over 2007–2010 are set against Curreli's Table 4 means and standard errors for the same communities. Agreement within two standard errors for the communities that both datasets populate admits the spring rows; disagreement is reported with the rows withheld, because it would mean the quadrats, the datum or the period differ from hers and the transfer of standing has no basis.
Outputs. A committed table, one row per community, metric and period, with the columns named above, written by a new pass of Script 26 alongside the annual-minimum series; a supplementary table generated from it; the path of the community file and the reproduction tolerance held in the configuration module; and a task-register row whose check reads the committed table. The values enter the report only through that table, and the spring-level counterpart of the quadrat comparison of report Section 4.8.3 could then be made.

S12.6 What it would and would not settle

It would give MSL5 the reference table it lacks, on Curreli's own quadrats and by her own method, so that a spring-level status could be stated for a well or a slack and the operational spring-level forecast could report it. It would not turn the values into viability limits: they would describe where each community was found on the metric, at one site, in one decade, with ranges that overlap — as Curreli's do — and they would inherit the small counts of the source, of the order of fifteen quadrats at two to six piezometers per community. Nor would it change the cluster-level readings of the report: a cluster mean averages dipwells at every topographic position, and remains deeper than any slack community by construction, so community status would be read at the quadrat or the slack floor and not at the cluster.

S12.7 Status

Specified 2026-09-21 under decision D-190; not built. The one outstanding input is the NVC community of each permanent quadrat by survey year (S12.4). The code, the datum offsets and the hydrological metrics are in place; the pass is a short addition to Script 26 once the column exists.